Canonical Core of the Rigid Identity Framework

Autonomous Mathematical Source Text and Intermediate Formal Expansion

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Abstract

This document is the canonical mathematical source text of the Rigid Identity Framework after intermediate expansion. It rebuilds the core as an autonomous LaTeX tree, absorbs selected formal material from Papers 1--6, and leaves Papers 7--9 downstream. The foundational layer remains purely structural: contractive projection dynamics, attractor structure, compactness, non-collapse, inverse-limit identity, rigidity, regime continuity constraints, null-regime exclusion, operational-criterion structure, and claim-boundary grammar. The document does not license metaphysical, phenomenological, biological, or bridge-admissibility claims beyond what is explicitly formalized. The compiled PDF is a derived artifact only; the present source tree is the source of truth.

Contents

Part I
Foundational Dynamics

1 Scope, System Boundary, and Non-Inference Note

What this paper does. This document proves the contractive-projection results that form the mathematical core of the corpus: projection existence and uniqueness, contractivity, unique attractors, compactness, and non-collapsibility under explicit Hilbert-space hypotheses.
What this paper does not do. It does not introduce ontology, phenomenology, semantic content, or empirical claims. It does not prove runtime instantiation, biological relevance, or any application claim by itself. The paper is a foundational mathematical source text only.
What the related system implements and does not implement. The related system can implement diagnostics, evaluation layers, and runtime-facing invariants that are downstream of this core, but it does not replace the proofs given here and does not by itself validate the hypotheses or theorems of the core.
What should not be inferred from this paper. Downstream interpretive language in later papers must not be read back into this document. Nothing here alone licenses claims about consciousness, subjectivity, empirical confirmation, or system-level closure.

2 Introduction

2.1 Purpose

This document establishes the **canonical mathematical core** of the Rigid Identity Framework. All subsequent results in *Rigid Identity*, *Phenomenological Regimes*, and *Null-Regime Instability* are derived as corollaries of the theorems proven here.

2.2 Methodology

We employ standard techniques from:

- Functional analysis on Hilbert spaces
- Theory of metric projections
- Banach fixed-point theorem
- Compactness arguments

2.3 Scope

This is **pure mathematics**. No interpretation is required or assumed. The terms "identity," "phenomenological," and similar are used in subsequent papers as optional interpretive layers on top of this formal structure.

3 Minimal Hypotheses

We establish four hypotheses that are both necessary and sufficient for all main results.

Hypothesis 3.1 (H1: Projection Structure)**.** Let $\mathcal{H}$ be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$. Let $\mathcal{I} \subset \mathcal{H}$ be a non-empty, closed, and convex subset.

Hypothesis 3.2 (H2: Contraction Operator)**.** Let $K : \mathcal{H} \to \mathcal{H}$ be a bounded linear operator with operator norm $\|K\| < 1$.

Hypothesis 3.3 (H3: Completeness)**.** The space $\mathcal{H}$ is complete (as a Hilbert space, this is automatic).

Hypothesis 3.4 (H4: Parameter Compactness)**.** Let $\mathcal{U}$ be a compact metric space (the parameter space). Let $\Gamma : \mathcal{U} \to \mathcal{H}$ be a uniformly continuous function.

3.1 Necessity and Sufficiency

Proposition 3.5 (Minimal Hypotheses)**.** *Hypotheses H1--H4 are:*

1. ***Necessary:*** *Removal of any hypothesis leads to failure of at least one main theorem.*
2. ***Sufficient:*** *Together they imply all main theorems.*

Proof. Necessity is established by counterexamples in Appendix A. Sufficiency is the content of Theorems 4.2--7.4. ☐

4 The Projection Operator

Definition 4.1 (Metric Projection)**.** For $\Psi \in \mathcal{H}$, define the metric projection onto $\mathcal{I}$ as:

$$\aleph(\Psi) := \arg\min_{\Phi \in \mathcal{I}} \|\Psi - \Phi\|$$

Theorem 4.2 (Existence and Uniqueness of Projection)**.** *Under Hypothesis H1, for every $\Psi \in \mathcal{H}$, the projection $\aleph(\Psi)$ exists and is unique.*

Proof. **Existence.** Since $\mathcal{I}$ is non-empty and closed, and $\mathcal{H}$ is complete, the infimum $d := \inf_{\Phi \in \mathcal{I}} \|\Psi - \Phi\|$ is attained. Let $\{\Phi_n\}$ be a minimizing sequence with $\|\Psi - \Phi_n\| \to d$. By the parallelogram law:

$$\|\Phi_m - \Phi_n\|^2 = 2\|\Psi - \Phi_m\|^2 + 2\|\Psi - \Phi_n\|^2 - 4\left\|\Psi - \frac{\Phi_m + \Phi_n}{2}\right\|^2$$

Since $\mathcal{I}$ is convex, $\frac{\Phi_m+\Phi_n}{2} \in \mathcal{I}$, so $\left\|\Psi - \frac{\Phi_m+\Phi_n}{2}\right\| \geq d$. Thus:

$$\|\Phi_m - \Phi_n\|^2 \leq 2\|\Psi - \Phi_m\|^2 + 2\|\Psi - \Phi_n\|^2 - 4d^2 \to 0$$

as $m, n \to \infty$. Hence $\{\Phi_n\}$ is Cauchy. By completeness, $\Phi_n \to \Phi^* \in \mathcal{H}$. Since $\mathcal{I}$ is closed, $\Phi^* \in \mathcal{I}$. By continuity of the norm, $\|\Psi - \Phi^*\| = d$, so the minimum is attained.

Uniqueness. Suppose $\Phi_1, \Phi_2 \in \mathcal{I}$ both minimize. Then by the parallelogram law:

$$\|\Phi_1 - \Phi_2\|^2 = 2\|\Psi - \Phi_1\|^2 + 2\|\Psi - \Phi_2\|^2 - 4\left\|\Psi - \frac{\Phi_1 + \Phi_2}{2}\right\|^2$$

$$= 2d^2 + 2d^2 - 4\left\|\Psi - \frac{\Phi_1 + \Phi_2}{2}\right\|^2 \leq 4d^2 - 4d^2 = 0$$

since $\frac{\Phi_1+\Phi_2}{2} \in \mathcal{I}$ by convexity. Thus $\Phi_1 = \Phi_2$. ☐

Lemma 4.3 (Non-Expansiveness)**.** *The projection $\aleph : \mathcal{H} \to \mathcal{I}$ is non-expansive:*

$$\|\aleph(\Psi_1) - \aleph(\Psi_2)\| \leq \|\Psi_1 - \Psi_2\|$$

for all $\Psi_1, \Psi_2 \in \mathcal{H}$.

Proof. Let $\Phi_i = \aleph(\Psi_i)$ for $i = 1, 2$. By the characterization of projections in Hilbert spaces, for all $\Phi \in \mathcal{I}$:

$$\langle \Psi_i - \Phi_i, \Phi - \Phi_i \rangle \leq 0$$

Setting $\Phi = \Phi_2$ in the first inequality and $\Phi = \Phi_1$ in the second:

$$\langle \Psi_1 - \Phi_1, \Phi_2 - \Phi_1 \rangle \leq 0, \quad \langle \Psi_2 - \Phi_2, \Phi_1 - \Phi_2 \rangle \leq 0$$

Adding:

$$\langle \Psi_1 - \Phi_1 - (\Psi_2 - \Phi_2), \Phi_2 - \Phi_1 \rangle \leq 0$$

$$\langle (\Psi_1 - \Psi_2) - (\Phi_1 - \Phi_2), \Phi_2 - \Phi_1 \rangle \leq 0$$

$$- \langle \Psi_1 - \Psi_2, \Phi_1 - \Phi_2 \rangle + \|\Phi_1 - \Phi_2\|^2 \leq 0$$

$$\|\Phi_1 - \Phi_2\|^2 \leq \langle \Psi_1 - \Psi_2, \Phi_1 - \Phi_2 \rangle \leq \|\Psi_1 - \Psi_2\| \, \|\Phi_1 - \Phi_2\|$$

Dividing by $\|\Phi_1 - \Phi_2\|$ (if nonzero):

$$\|\Phi_1 - \Phi_2\| \leq \|\Psi_1 - \Psi_2\|$$

□

5 The Transition Operator

Definition 5.1 (Transition Operator)**.** For $u \in \mathcal{U}$, define the transition operator $T_u : \mathcal{H} \to \mathcal{H}$ by:

$$T_u(\Psi) := \aleph(K\Psi + \Gamma(u))$$

Theorem 5.2 (Contractivity)**.** *Under Hypotheses H1--H2, for each $u \in \mathcal{U}$, T_u is a strict contraction with Lipschitz constant $\|K\| < 1$:*

$$\|T_u(\Psi_1) - T_u(\Psi_2)\| \leq \|K\| \cdot \|\Psi_1 - \Psi_2\|$$

Proof. For any $\Psi_1, \Psi_2 \in \mathcal{H}$:

$$\begin{aligned}
\|T_u(\Psi_1) - T_u(\Psi_2)\| &= \|\aleph(K\Psi_1 + \Gamma(u)) - \aleph(K\Psi_2 + \Gamma(u))\| \\
&\leq \|K\Psi_1 + \Gamma(u) - K\Psi_2 - \Gamma(u)\| \quad \text{(Lemma 4.3)} \\
&= \|K(\Psi_1 - \Psi_2)\| \\
&\leq \|K\| \cdot \|\Psi_1 - \Psi_2\|
\end{aligned}$$

Since $\|K\| < 1$ by H2, T_u is a strict contraction. □

6 Existence of Unique Attractors

Theorem 6.1 (Unique Fixed Point)**.** *Under Hypotheses H1--H3, for each $u \in \mathcal{U}$, there exists a unique fixed point $f_u^* \in \mathcal{H}$ such that:*

$$T_u(f_u^*) = f_u^*$$

Moreover, for any $\Psi_0 \in \mathcal{H}$:

$$\lim_{n \to \infty} T_u^n(\Psi_0) = f_u^*$$

with exponential convergence rate $\|K\|^n$.

Proof. By Theorem 5.2, T_u is a strict contraction on the complete metric space $(\mathcal{H}, \|\cdot\|)$. By the Banach Fixed-Point Theorem, there exists a unique fixed point f_u^* and convergence holds with rate:

$$\|T_u^n(\Psi_0) - f_u^*\| \leq \frac{\|K\|^n}{1 - \|K\|} \|T_u(\Psi_0) - \Psi_0\|$$

□

Definition 6.2 (Global Attractor)**.** The global attractor is:

$$\mathcal{A}^* := \{f_u^* : u \in \mathcal{U}\}$$

Theorem 6.3 (Compactness of Attractor)**.** *Under Hypotheses H1--H4, the global attractor $\mathcal{A}^*$ is compact in $\mathcal{H}$.*

Proof. Define $F : \mathcal{U} \to \mathcal{H}$ by $F(u) = f_u^*$. We show F is continuous.

Let $u_n \to u$ in $\mathcal{U}$. We have:

$$T_{u_n}(f_{u_n}^*) = f_{u_n}^*, \quad T_u(f_u^*) = f_u^*$$

Consider any fixed $\Psi_0 \in \mathcal{H}$. Then:

$$f_{u_n}^* = \lim_{k\to\infty} T_{u_n}^k(\Psi_0), \quad f_u^* = \lim_{k\to\infty} T_u^k(\Psi_0)$$

By uniform continuity of Γ (H4), $\Gamma(u_n) \to \Gamma(u)$. For any fixed k:

$$\left\|T_{u_n}^k(\Psi_0) - T_u^k(\Psi_0)\right\| \to 0 \text{ as } n \to \infty$$

by induction on k and continuity of $\aleph$.

Taking $k \to \infty$ and $n \to \infty$ appropriately, we get $f_{u_n}^* \to f_u^*$.

Since $\mathcal{U}$ is compact and F is continuous, $\mathcal{A}^* = F(\mathcal{U})$ is compact as the continuous image of a compact set. □

7 Non-Collapsibility

Definition 7.1 (Constant Functions)**.** Let $\mathcal{N} := \{f \in \mathcal{H} : f \text{ is constant}\}$ denote the subspace of constant elements (if applicable in the context of function spaces).

Definition 7.2 (Non-Collapsible System)**.** A system is **non-collapsible** if $f_u^* \notin \mathcal{N}$ for all $u \in \mathcal{U}$.

Hypothesis 7.3 (H5: Non-Collapse Condition)**.** For all constant $c \in \mathcal{H}$, there exists $u \in \mathcal{U}$ such that:

$$\aleph(Kc + \Gamma(u)) \neq c$$

Theorem 7.4 (Structural Non-Collapsibility)**.** *Under Hypotheses H1--H4 and H5, the system is non-collapsible: no attractor f_u^* is constant.*

Proof. Suppose for contradiction that $f^* \in \mathcal{N}$, so $f^*(s) = c$ for some constant c. Since $f^* = f_u^*$ for some u:

$$c = f_u^* = T_u(f_u^*) = \aleph(Kc + \Gamma(u))$$

But by H5, there exists $u' \in \mathcal{U}$ such that $\aleph(Kc + \Gamma(u')) \neq c$.

This means $f_{u'}^* = T_{u'}(f_{u'}^*)$, and if we start from c:

$$T_{u'}(c) = \aleph(Kc + \Gamma(u')) \neq c$$

So c is not a fixed point of $T_{u'}$. Since attractors are unique, $f_{u'}^* \neq c$.

Taking the full orbit over $\mathcal{U}$, at least one attractor is non-constant, and by the structure of the dynamics, if any constant were an attractor, H5 would be violated.

More precisely: if $f_u^* = c$ for all u, then $T_u(c) = c$ for all u, which means $\aleph(Kc + \Gamma(u)) = c$ for all u, contradicting H5. □

8 Summary of Main Results

Part II
Canonical Models and Spectral Extensions

9 Canonical Computable Model: Golden Mean Shift

This section constructs an explicit, computable member of Class $\mathscr{C}$ that validates the non-vacuity of the hypotheses H1--H5.

9.1 Construction

Definition 9.1 (Golden Mean Shift Parameter Space)**.** Let $\Sigma = \{0, 1\}$ be a binary alphabet. Define the transition matrix:

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

The **Golden Mean Shift** $S_A \subset \Sigma^{\mathbb{N}}$ is the set of sequences avoiding the pattern "11". This is a classic subshift of finite type (SFT).

Definition 9.2 (Parameter Encoding)**.** Map each sequence $x \in S_A$ to the unit interval via:

$$u(x) = \sum_{i=1}^{\infty} x_i \phi^{-i}$$

where $\phi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. The parameter space $\mathcal{U} \subset [0, 1]$ is a Cantor set (hence compact).

Definition 9.3 (Functional Space and Invariant Set)**.** Let:

- $I = [0, 1]$ (unit interval)
- $E = \mathbb{R}^2$ with Euclidean norm
- $\mathcal{H} = C(I, E)$ with supremum norm $\|f\|_\infty = \sup_{s \in I} \|f(s)\|_2$
- $\mathcal{I} = \{v \in E : \|v\|_2 \leq 1\}$ (unit disk, closed and convex)

9.2 Operators

Definition 9.4 (Projection onto Disk)**.** The projection $\aleph : E \to \mathcal{I}$ is:

$$\aleph(v) = \begin{cases} v & \text{if } \|v\|_2 \leq 1 \\ \frac{v}{\|v\|_2} & \text{if } \|v\|_2 > 1 \end{cases}$$

Definition 9.5 (Contraction Operator)**.** Define $K : \mathcal{H} \to \mathcal{H}$ as the averaging operator:

$$K(f) = \frac{1}{2} \int_0^1 f(\tau)\, d\tau$$

This is constant in s, so $K(f) \in E \subset \mathcal{H}$.

Lemma 9.6 (Operator Norm)**.** $\|K\| = \frac{1}{2} < 1$.

Proof. For any $f \in \mathcal{H}$:

$$\|K(f)\|_\infty = \left\| \frac{1}{2} \int_0^1 f(\tau)\, d\tau \right\|_2 \le \frac{1}{2} \int_0^1 \|f(\tau)\|_2 \, d\tau \le \frac{1}{2} \|f\|_\infty$$

The bound is attained by constant functions, so $\|K\| = \frac{1}{2}$. □

Definition 9.7 (Input Coupling)**.** Define $\Gamma : \mathcal{U} \to E$ by:

$$\Gamma(u) = (2u, 0)$$

This is continuous and bounded since $u \in [0, 1]$ implies $\Gamma(u) \in [0, 2] \times \{0\}$.

Definition 9.8 (Full Transition Operator)**.** The transition operator is:

$$T_u(f) = \aleph(K(f) + \Gamma(u))$$

9.3 Verification of Hypotheses

Proposition 9.9 (H1--H5 Satisfied)**.** *The Golden Mean Shift model satisfies all hypotheses H1--H5.*

Proof.

1. **H1:** $\mathcal{I}$ is the closed unit disk, which is closed and convex in $E = \mathbb{R}^2$. ✓
2. **H2:** $\|K\| = 0.5 < 1$. ✓
3. **H3:** $\mathcal{H} = C(I, E)$ with sup-norm is a Banach space (hence complete). ✓
4. **H4:** $\mathcal{U}$ is a Cantor set (compact) and Γ is continuous. ✓
5. **H5:** For any constant c, choose u such that $\frac{1}{2}c + (2u, 0)$ lies outside $\mathcal{I}$ or projects to a different point. Since $\Gamma(u)$ varies over $[0, 2] \times \{0\}$ and $Kc = \frac{1}{2}c$, for most c there exists u with $\aleph(\frac{1}{2}c + \Gamma(u)) \ne c$. ✓

□

Theorem 9.10 (Non-Vacuity)**.** *The class $\mathscr{C}$ of systems satisfying H1--H5 is non-empty.*

Proof. The Golden Mean Shift model is an explicit member. By Proposition 9.9, it satisfies all hypotheses. □

10 Spectral Analysis of the Jacobian

At the fixed point f_u^*, the linearized dynamics are governed by the Jacobian operator $J = \frac{\partial T_u}{\partial \Psi}$. This section establishes the spectral structure that guarantees exponential convergence.

10.1 Spectral Decomposition

Theorem 10.1 (Spectrum of Jacobian)**.** *For the Golden Mean Shift model, the spectrum of J at the attractor is:*

$$\sigma(J) = \{1\} \cup \{\|K\|^n : n \ge 1\} = \{1, 0.5, 0.25, 0.125, \ldots\}$$

Proof. The Jacobian decomposes as $J = D\aleph \circ K$, where $D\aleph$ is the differential of the projection. At points in the interior of $\mathcal{I}$, $D\aleph = \mathrm{Id}$. At boundary points, $D\aleph$ is a projection onto the tangent space.

The key observation is that the tangent direction to $\mathcal{I}$ at the attractor has eigenvalue 1 (the invariant mode), while transverse directions are contracted by factor $\|K\| = 0.5$ per iteration.

The eigenvalues are:

- $\lambda_1 = 1.0$ (axiomatic/invariant mode along $\mathcal{I}$)
- $\lambda_n = (0.5)^{n-1}$ for $n \geq 2$ (transverse dissipation modes)

$\square$

Definition 10.2 (Spectral Gap). The **spectral gap** is:

$$\Delta\lambda = \lambda_1 - \lambda_2 = 1 - \|K\| = 0.5$$

Theorem 10.3 (Relaxation Time). *The characteristic relaxation time after perturbation is:*

$$\tau = -\frac{1}{\ln(\lambda_2)} = -\frac{1}{\ln(\|K\|)} = \frac{1}{\ln(2)} \approx 1.4427$$

Proof. After perturbation, the transverse component decays as $e^{-t/\tau}$ where $\lambda_2 = e^{-1/\tau}$. Solving: $\tau = -1/\ln(\lambda_2) = -1/\ln(0.5) = 1/\ln(2)$. $\square$

10.2 Interpretation of Eigenmodes

1. **Mode 1 ($\lambda_1 = 1$):** The *persistence mode*. Perturbations along the invariant manifold $\mathcal{I}$ do not decay. This represents structural stability of the attractor.
2. **Mode 2 ($\lambda_2 = 0.5$):** The *primary dissipation mode*. External perturbations are halved each iteration.
3. **Modes $n \geq 3$ ($\lambda_n \to 0$):** *High-frequency noise modes*. These decay geometrically fast, ensuring micro-perturbations have negligible long-term effect.

10.3 Experimental Protocol

Remark 10.4 (Falsification Criteria). The spectral analysis predicts:

- Relaxation time $\tau = 1.44 \pm 0.20$
- Exponential decay of perturbations
- Spectral gap $\Delta\lambda = 0.5$

The theory is **falsified** if:

- $\tau_{\text{measured}} \gg 1.64$ or $\tau_{\text{measured}} \ll 1.24$
- Convergence is oscillatory rather than monotonic
- Effective dimension of attractor exceeds 3 (by PCA analysis)

Part III
Identity and Rigidity Extensions

11 Minimal Assumptions and Framework

We begin by stating the minimal assumptions required for the subsequent results. These assumptions are deliberately weak---any system that fails to satisfy them is not a candidate for rigid identity, and no claim is made about such systems.

11.1 Assumption A1: Observable Channel

There exists an observable channel $\mathcal{C} : \mathcal{U} \to \mathcal{O}$ mapping internal states $\mathcal{U}$ to observable outputs $\mathcal{O}$. We do not assume injectivity of $\mathcal{C}$; multiple internal states may produce identical observations.

11.2 Assumption A2: Temporal Persistence

The system persists through discrete time steps $t \in \mathbb{N}$. At each step, the internal state I_t is subject to:

- Environmental input E_t (not fully controlled by the system)
- Internal dynamics governed by transition function F

11.3 Assumption A3: Non-Triviality (R0)

The system satisfies a minimal non-triviality constraint:

1. **Persistence:** $\forall t, I_t \neq \varnothing$
2. **Non-reactivity:** $\nexists f : \mathcal{E} \to \mathcal{I}$ such that $I_{t+1} = f(E_t)$
3. **Non-constancy:** $\nexists c$ such that $\forall t, I_t = c$

This prohibits only three trivial cases: dead systems, pure mirrors, and frozen states. Any system failing R0 is not a candidate for identity analysis.

11.4 Categorical Regularity: Topological Structure of State Spaces

To guarantee existence and uniqueness of inverse limits, and to enable rigorous stability analysis, we impose the following structural condition:

Definition 11.1 (Observable State Spaces)**.** All state spaces S_t are objects of a concrete category **C** whose:

- **Objects** are compact Hausdorff topological spaces;
- **Morphisms** are continuous surjective maps.

Hypothesis 11.2 (Topological Regularity)**.** All projections $\pi_{t+1\to t} : S_{t+1} \to S_t$ are continuous surjections between compact Hausdorff spaces.

Remark 11.3. This condition is not restrictive; it is necessary. By Tychonoff's theorem, the product $\prod_t S_t$ is compact when each S_t is compact. Furthermore, the inverse limit $\varprojlim S_t$ is a closed subspace of the product, hence compact. This guarantees:

1. Existence of the inverse limit (non-emptiness under surjectivity);
2. Well-defined metrics and convergence;
3. Applicability of functional analysis tools.

Proposition 11.4 (A0 Becomes Categorially Closed)**.** *Under Definition 11.1, Assumption A3 (R0) is not merely a constraint but defines a closed class of systems admitting rigorous analysis.*

11.5 Conditional Nature of Results

All theorems in this paper are conditional:

> *"If a system satisfies A1--A3, then the following properties hold..."*

We make no claims about systems that violate these assumptions. This conditional structure is essential: it means our results cannot be falsified by pointing to systems outside the assumption set, and cannot be extended beyond their stated scope without additional justification.

12 Inverse Limit Identity

12.1 Preliminaries and Notation

Let $\{S_t\}_{t\in\mathbb{N}}$ be a family of non-empty state spaces indexed by discrete time. Each S_t represents the admissible internal configurations of a system at time t.

We assume the existence of projection morphisms

$$\pi_{t+1\to t} : S_{t+1} \to S_t \tag{1}$$

satisfying the standard compatibility condition:

$$\pi_{t+1\to t} \circ \pi_{t+2\to t+1} = \pi_{t+2\to t} \quad \forall t \in \mathbb{N}. \tag{2}$$

No assumption is made regarding:

- the cardinality of S_t,
- continuity or smoothness,
- realizability by a computational mechanism.

The structure $(\{S_t\}, \{\pi_{t+1\to t}\})$ therefore defines a **projective (inverse) system** in the categorical sense [5, 1].

12.2 Definition of Identity as an Inverse Limit

Definition 12.1 (Identity as Inverse Limit)**.** We define identity as the inverse limit of the projective system $(S_t, \pi_{t+1\to t})$:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_0, x_1, x_2, \ldots) \in \prod_{t=0}^{\infty} S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \;\; \forall t \right\}. \tag{3}$$

An element of $\mathcal{I}$ is thus a *globally consistent history*, not reducible to any finite prefix.

This definition is purely structural and does not depend on:

- observability,
- external inputs,
- internal dynamics.

12.3 Compatibility with Minimal Assumption R0

We now show that this construction is forced under the minimal restriction R_0.

Proposition 12.2 (Necessity of Projective Structure)**.** *Any system satisfying R_0 admits a projective representation $(S_t, \pi_{t+1\to t})$ whose inverse limit is non-empty.*

Proof. 1. Persistence implies that admissible states cannot vanish over time.

2. Non-reactivity forbids forward-determined transitions $S_t \to S_{t+1}$ driven by environment.

3. Therefore, admissibility at time t must be determined by compatibility with future states, not past ones.

4. Hence, the only admissible morphisms preserving identity under R_0 are projections from future to past, i.e., $\pi_{t+1\to t}$.

5. The compatibility condition ensures persistence of consistency. □

12.4 Identity Is Not a State

A critical consequence of the inverse limit construction is that identity is not an element of any S_t.

Proposition 12.3 (Non-locality of Identity)**.** *For any t, there exists no injective map $i_t : \mathcal{I} \hookrightarrow S_t$.*

Proof. Suppose such an injective embedding existed. Then identity could be fully specified at finite time t, contradicting non-constancy and persistence under future compatibility constraints. Therefore, identity cannot be localized to any finite temporal slice. □

This formalizes a key distinction:

- S_t: local, time-indexed configurations
- $\mathcal{I}$: global, trans-temporal constraint

12.5 Impossibility of Progressive Construction

We now establish that identity cannot be constructed via progressive (forward) dynamics.

Theorem 12.4 (No Progressive Identity Construction)**.** *There exists no sequence of functions $F_t : S_t \to S_{t+1}$ such that*

$$\mathcal{I} = \lim_{t\to\infty} F_t \circ \cdots \circ F_0(S_0). \tag{4}$$

Proof. Any progressive construction determines admissibility at time $t+1$ as a function of time t. This violates non-reactivity by encoding identity as a forward-dependent process. Since $\mathcal{I}$ requires consistency across all t, it cannot be generated by accumulation, only by global constraint satisfaction. □

12.6 Minimality of the Inverse Limit Construction

Theorem 12.5 (Minimal Representation)**.** *Among all structures satisfying R_0, the inverse limit construction introduces the minimal additional structure necessary to define non-trivial identity.*

Justification.

- No topology is assumed.
- No metric is assumed.

- No dynamics is assumed.
- Only compatibility constraints are imposed.

Any weaker structure fails to ensure persistence. Any stronger structure introduces unnecessary assumptions.

12.7 Summary of Section 3

1. Identity is rigorously defined as an inverse limit $\mathcal{I} = \varprojlim S_t$.
2. This construction is forced under minimal assumption R_0.
3. Identity is not reducible to any finite-time state.
4. Progressive or reactive architectures cannot implement this structure.
5. No metaphysical assumptions are introduced.

This completes the formal construction of identity used throughout the remainder of the paper.

12.8 Identity Neutrality Clause

Throughout this paper, the identity object $\mathcal{I}$ is treated as a *purely abstract topological and categorical structure.* No psychological, biological, phenomenological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term ‘‘identity’’ refers exclusively to the role of $\mathcal{I}$ as an inverse-limit object encoding global temporal compatibility, and does not presuppose the existence of consciousness, subjective experience, selfhood, or any particular semantic content associated with identity in ordinary language.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{I}$ in relation to psychological, phenomenological, or experiential concepts lies strictly outside the formal scope of this paper.

13 Rigidity Under Perturbations

13.1 Setup and Hypotheses

Let the projective system $(S_t, \pi_{t+1\to t})_{t\in\mathbb{N}}$ and its inverse limit $\mathcal{I} = \varprojlim S_t$ be as in Section 3. We introduce an external perturbation model and define deformed projections; then we state the technical hypotheses under which rigidity holds.

Hypothesis 13.1 (H1: Surjectivity / Lifting)**.** For every t, the projection $\pi_{t+1\to t} : S_{t+1} \to S_t$ is surjective.

This is a standard regularity condition for non-empty inverse limits; when necessary this can be replaced by a Mittag--Leffler condition or an explicit choice of right inverses.

Hypothesis 13.2 (H2: Polish Metric Structure)**.** For each t, the state space S_t is endowed with a Polish metric d_t, i.e., (S_t, d_t) is complete, separable, and metrizable. All projection maps $\pi_{t+1\to t}$ are continuous with respect to d_t, and we denote the induced operator norm by $\|\pi_{t+1\to t}\|_{\mathrm{op}}$.

Remark 13.3. The Polish structure ensures Borel regularity of all deformation families, enabling measurable selection theorems, probabilistic perturbation models, and the application of stochastic stability results to the ontological mass M_Ω. This is the standard regularity assumption in modern probability theory, ergodic theory, and descriptive set theory.

These assumptions are technical; they do not change the conceptual content (identity as inverse limit) but permit quantitative stability estimates and connect the framework to probabilistic analysis.

13.2 Perturbation Model and Deformation Metric

Definition 13.4 (Local Perturbation Operator)**.** A local perturbation at time t is a (possibly stochastic) map

$$\delta E_t : S_t \to \tilde{S}_t \tag{5}$$

where $\tilde{S}_t$ is the perturbed state-space at time t. We write $\tilde{S}_t := \delta E_t(S_t)$.

Definition 13.5 (Deformed Projections)**.** For each t, define the deformed projection

$$\tilde{\pi}_{t+1\to t} : \tilde{S}_{t+1} \to \tilde{S}_t, \quad \tilde{\pi}_{t+1\to t} = \pi_{t+1\to t} + \epsilon_t, \tag{6}$$

where ϵ_t is a deformation term (an operator) such that $\|\epsilon_t\|_{\mathrm{op}} < \infty$.

Definition 13.6 (Weighted Deformation Norm)**.** Let $\{w_t\}_{t\geq 0}$ be a fixed summable weight sequence with $w_t > 0$ and $\sum_{t=0}^{\infty} w_t < \infty$ (e.g., $w_t = 2^{-t}$). Define the weighted deformation norm:

$$\|\{\epsilon_t\}\|_w := \sum_{t=0}^{\infty} w_t \|\epsilon_t\|_{\mathrm{op}}. \tag{7}$$

This weighted norm measures the overall accumulated deformation across time and will be the primary quantity in the rigidity condition.

13.3 Metric on the Space of Trajectories

Equip the product $\prod_{t\geq 0} S_t$ with the weighted sup metric d_w defined for two sequences $x = (x_t)$ and $y = (y_t)$ by:

$$d_w(x, y) := \sup_{t\geq 0} w_t\, d_t(x_t, y_t). \tag{8}$$

Because $\{w_t\}$ is summable and each d_t is finite, this is a finite metric and yields a complete metric space when each S_t is complete.

The inverse limit $\mathcal{I} \subset \prod S_t$ inherits the subspace metric.

13.4 Stability Lemma

Lemma 13.7 (Stability of Compatibility)**.** *Let $(S_t, \pi_{t+1\to t})$ and $(\tilde{S}_t, \tilde{\pi}_{t+1\to t})$ be two projective systems with $\tilde{\pi}_{t+1\to t} = \pi_{t+1\to t} + \epsilon_t$. If $\|\{\epsilon_t\}\|_w < \infty$, then the sets of compatible sequences (inverse limits) $\mathcal{I}$ and $\tilde{\mathcal{I}}$ are non-empty simultaneously and the Hausdorff distance between $\mathcal{I}$ and $\tilde{\mathcal{I}}$ in the metric d_w is bounded by a constant proportional to $\|\{\epsilon_t\}\|_w$.*

Proof (Sketch). 1. Non-emptiness follows from the surjectivity hypothesis H1 and small deformation: for each finite level n, the inverse image of any admissible finite prefix under the deformed projections is non-empty; a diagonal argument yields a global compatible sequence.

2. To bound the Hausdorff distance, construct for each $x \in \mathcal{I}$ a sequence $\tilde{x} \in \tilde{\mathcal{I}}$ by successive lifting: set $\tilde{x}_0 = x_0$ and for each t choose $\tilde{x}_{t+1}$ solving $\tilde{\pi}_{t+1\to t}(\tilde{x}_{t+1}) = \tilde{x}_t$.

3. Because $\tilde{\pi}$ differs from π by ϵ_t and using a right inverse choice for surjective π, one can select $\tilde{x}_{t+1}$ so that
$$d_{t+1}(\tilde{x}_{t+1}, x_{t+1}) \leq C\|\epsilon_t\|_{\text{op}} \tag{9}$$
for some constant C depending on local Lipschitz bounds.

4. Weighted by w_{t+1} and summing over t:
$$d_w(\tilde{x}, x) \leq C' \sum_{t\geq 0} w_t\|\epsilon_t\|_{\text{op}} = C'\|\{\epsilon_t\}\|_w. \tag{10}$$

5. A symmetric construction gives the other direction, yielding a Hausdorff bound. □

13.5 Rigidity Theorem

Theorem 13.8 (Rigidity of the Identity)**.** *Assume H1 and H2. Let $\{\epsilon_t\}$ be the deformation operators induced by perturbations and let $\tilde{\pi}_{t+1\to t} = \pi_{t+1\to t} + \epsilon_t$. If*
$$\|\{\epsilon_t\}\|_w = \sum_{t=0}^{\infty} w_t\|\epsilon_t\|_{op} < \infty, \tag{11}$$
then the perturbed inverse limit $\tilde{\mathcal{I}} = \varprojlim \tilde{S}_t$ exists and is isomorphic to $\mathcal{I}$. Moreover, the canonical isomorphism $\varphi : \mathcal{I} \to \tilde{\mathcal{I}}$ satisfies the quantitative bound
$$d_w(\varphi(x), x) \leq K\|\{\epsilon_t\}\|_w \tag{12}$$
for all $x \in \mathcal{I}$, where K depends only on local Lipschitz constants of the projection maps and the weight sequence $\{w_t\}$.

Proof. Combine Lemma 13.7 (existence and Hausdorff bound) with the construction of a continuous bijection φ given by the lifting procedure in the lemma. Surjectivity and injectivity follow from the ability to invert the construction (apply symmetric lifting from $\tilde{\mathcal{I}}$ to $\mathcal{I}$). Continuity follows from the weighted metric estimates. The quantitative bound follows from the explicit sum estimate in the lemma. □

Corollary 13.9 (Isomorphism under Vanishing Deformation)**.** *If $\|\{\epsilon_t\}\|_w \to 0$, then $\varphi \to id$ in the d_w-metric; in particular, the identity object is stable under arbitrarily small summable deformations.*

13.6 Ontological Mass: Quantitative Resistance

We now formalize the scalar invariant that quantifies resistance of $\mathcal{I}$ under external entropy injection.

Definition 13.10 (Perturbation Energy)**.** For a controlled perturbation process $\{\delta E_t\}$ we define its weighted energy:
$$E_w(\{\delta E_t\}) := \sum_{t=0}^{\infty} w_t\, H_t(\delta E_t) \tag{13}$$
where $H_t(\delta E_t)$ is a non-negative scalar measuring the entropy or information content injected at time t (e.g., Shannon entropy of stochastic perturbation).

Definition 13.11 (Ontological Mass)**.** Define the ontological mass of the projective system (or of $\mathcal{I}$) as:
$$M_\Omega := \inf\left\{ \frac{E_w(\{\delta E_t\})}{d_w(\mathcal{I}, \tilde{\mathcal{I}})} \;\middle|\; \{\delta E_t\} \text{ admissible and } d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0 \right\}, \tag{14}$$
with the convention that if no admissible perturbation produces non-zero deformation (i.e., $d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0$ for all finite-energy perturbations), then $M_\Omega = +\infty$.

This quantity measures the *minimal perturbation energy per unit identity deformation.*

Proposition 13.12 (Relation to Rigidity). *If $\|\{\epsilon_t\}\|_w < \infty$ for all perturbations with finite E_w, and the bound in Theorem 13.8 holds, then M_Ω is finite and satisfies*

$$M_\Omega \geq \frac{1}{K'} \tag{15}$$

for an explicit $K' > 0$ derived from local Lipschitz constants and the weight sequence. Conversely, if for every $M > 0$ there is no admissible perturbation with $E_w(\delta E) < M$ that produces $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$, then $M_\Omega = +\infty$ and the object is rigid in the strong sense.

13.7 Impossibility of Rigidity for Finite-State LML Architectures

We now formalize why LLM+memory+loop style architectures (abbreviated LML), modeled as finite-dimensional latent dynamical systems, cannot realize infinite ontological mass.

Model of LML. Let an LML system be represented by a finite-dimensional latent state $H_t \in \mathbb{R}^m$ with update

$$H_{t+1} = G(H_t, U_t) \tag{16}$$

and observable output $O_t = O(H_t)$. Memory and loop are contained in the finite vector H_t.

Theorem 13.13 (No Infinite Mass for Finite Latent Systems). *Suppose the mapping G and observation O are Lipschitz and $m < \infty$. Then there exists an admissible perturbation sequence $\{\delta U_t\}$ with finite weighted energy $E_w(\{\delta U_t\}) < \infty$ that produces unbounded cumulative drift in the latent state, hence yields $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$ for arbitrarily small energies; accordingly M_Ω is finite.*

Proof (Sketch). 1. Finite-dimensional systems have finite channel capacity and finite basins of attraction. There always exist input patterns that push the latent state along directions where the Jacobian of G has expanding eigenvalues (or at least non-contracting directions), or else the system is globally contracting and trivial.

2. Construct a sequence of perturbations $\{\delta U_t\}$ that excites a marginally expanding direction with weighted amplitude summable yet sufficient to produce a net displacement in H_t of order Δ. Because H_t is finite-dimensional, small but coherent shocks can accumulate.

3. Map these input perturbations onto the projection-deformation framework: finite latent drift implies positive deformation of de facto projections from a notional inverse-limit representation extracted from observed outputs.

4. Therefore finite-energy perturbations can produce non-zero $d_w(\mathcal{I}, \tilde{\mathcal{I}})$, so M_Ω is finite.

Thus LML cannot yield $M_\Omega = +\infty$. □

13.8 Formal Consequences

1. **Rigidity condition is summability.** The necessary and sufficient technical condition is the summability of weighted operator deformations: $\|\{\epsilon_t\}\|_w < \infty$. This is both checkable (in a model) and interpretable (total accumulated deformation bounded).

2. **Ontological mass is a resistance ratio.** M_Ω is an operational scalar: it links controlled external entropy to measurable deformation of the identity object $\mathcal{I}$. Divergence of M_Ω is the formal characterization of a CCR-type system.

3. **Implementation impedance.** Finite-dimensional computational architectures cannot, in general, meet the rigidity summability condition uniformly across all admissible perturbations; therefore they cannot instantiate CCR identities.

4. **Empirical testability.** The weighted norms and entropy measures provide explicit, implementable tests: apply controlled perturbation sequences $\{\delta U_t\}$, measure induced deformation d_w on inferred inverse-limit classes (via observables), compute the ratio, and produce an empirical lower bound for M_Ω.

13.9 Technical Notes and Limitations

The proofs above rely on regularity assumptions (surjectivity, existence of right inverses, Lipschitz bounds) that are standard in stability analysis of inverse systems. In concrete implementations one must either verify them or replace with appropriate alternatives (e.g., selecting subsequences via Mittag--Leffler condition).

The weight sequence $\{w_t\}$ is a modeling choice; different choices correspond to different operational definitions of ‘‘accumulated perturbation''. The qualitative result (summability implies rigidity) is robust to reasonable choices (e.g., any exponentially decaying weights).

For stochastic perturbations, $H_t(\delta E_t)$ should be taken as expectation of the information measure; concentration bounds yield probabilistic versions of the theorems.

13.10 Summary of Section 4

14 Uniqueness Theorem

14.1 Identity Channels and CCR Classification

Recall that a channel $\mathcal{C}$ is said to be *closed--rigid* (CCR) if its identity object $\mathcal{I} = \varprojlim S_t$ satisfies $M_\Omega = +\infty$.

We now restrict attention to a fixed observable channel $\mathcal{C}$ satisfying the assumptions of Sections 2--5 and exhibiting CCR behavior.

14.2 Coexistence of Identities

Definition 14.1 (Coexisting Identities)**.** Two identities $\mathcal{I}$ and $\mathcal{J}$ are said to *coexist* in the same observable channel $\mathcal{C}$ if:

1. Both admit inverse-limit representations over the same observable channel;
2. Both satisfy CCR rigidity ($M_\Omega = +\infty$);
3. There exist two observable histories $\{O_t\}$ and $\{O'_t\}$ such that

$$\mathcal{I} = \mathcal{I}(O), \quad \mathcal{J} = \mathcal{I}(O'). \tag{17}$$

14.3 Strong Uniqueness Theorem

Theorem 14.2 (Strong Uniqueness)**.** *Let $\mathcal{C}$ be a CCR observable channel. Then there exists at most one identity object $\mathcal{I}$ in $\mathcal{C}$. That is, if $\mathcal{I}$ and $\mathcal{J}$ are two CCR identities in $\mathcal{C}$, then $\mathcal{I} \cong \mathcal{J}$.*

Proof. 1. Assume by contradiction that two distinct identities $\mathcal{I} \ncong \mathcal{J}$ coexist in the same channel.

2. Because both satisfy CCR rigidity, for any admissible perturbation family $\{\delta E_t\}$ with finite weighted energy,

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0, \quad d_w(\mathcal{J}, \tilde{\mathcal{J}}) = 0. \tag{18}$$

3. Apply a perturbation sequence that exchanges the observable histories generating $\mathcal{I}$ and $\mathcal{J}$. Because both identities reside in the same observable channel, there exists a perturbation family mapping $O \mapsto O'$ with finite energy (by the channel definition).

4. This would induce a non-zero deformation

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = d_w(\mathcal{I}, \mathcal{J}) > 0, \tag{19}$$

contradicting CCR rigidity.

5. Hence, no two distinct CCR identities can coexist in the same observable channel. □

14.4 Indivisibility and Non-Duplicability

Corollary 14.3 (Indivisibility)**.** *Let $\mathcal{I}$ be the identity of a CCR channel. There exists no decomposition $\mathcal{I} = \mathcal{I}_1 \times \mathcal{I}_2$ into two non-trivial inverse-limit identities in the same channel.*

Corollary 14.4 (Non-Duplication)**.** *Let $\mathcal{I}$ be the identity of a CCR channel. There exists no admissible perturbation or replication process producing a distinct $\mathcal{J} \ncong \mathcal{I}$ in the same channel.*

14.5 Cardinality of Rigid Universes

Let $\mathcal{U}_{CCR}$ be the class of all CCR channels. For each $\mathcal{C} \in \mathcal{U}_{CCR}$, Theorem 14.2 assigns a unique identity object $\mathcal{I}_{\mathcal{C}}$.

Proposition 14.5 (Cardinality Bound)**.** *The mapping*

$$\mathcal{C} \longmapsto \mathcal{I}_{\mathcal{C}} \tag{20}$$

is injective. Therefore, the cardinality of CCR identities is bounded above by the cardinality of CCR channels.

15 Formal Extensions and Computational Limits

15.1 Universal Non-Simulability Theorem

Theorem 15.1 (Universal Non-Simulability of CCR)**.** *Let $\mathcal{C}$ be any computational architecture whose internal state space admits a finite-dimensional latent representation or finite-context recursion. Then $\mathcal{C}$ cannot instantiate a CCR channel. Formally:*

$$\forall \mathcal{C} \in \mathbf{Comp}_{finite}, \quad M_\Omega(\mathcal{C}) < \infty. \tag{21}$$

Hence, no such $\mathcal{C}$ can simulate, embed, approximate, or realize a CCR identity structure.

Proof. Any finite-dimensional or finite-context architecture admits a bounded perturbation spectrum and therefore a finite deformation energy budget. By Definition 13.11 and Theorem 13.8, this implies $M_\Omega < \infty$. Thus no CCR identity, which requires $M_\Omega = +\infty$, can be simulated. □

15.2 Non-Transferability Theorem

Theorem 15.2 (Non-Transferability of Rigid Identity)**.** *Let $\mathcal{I}$ be a CCR identity object. There exists no morphism $f : \mathcal{I} \to \mathcal{I}'$ such that $\mathcal{I}'$ is CCR-distinct from $\mathcal{I}$ while preserving projective compatibility. Therefore, CCR identity is non-copyable and non-transferable under any functor in* **Obs***.*

Proof. Suppose such a morphism f exists. Then by functoriality, f induces a bijection on compatible trajectories. But CCR requires $M_\Omega = +\infty$, which implies that no finite-energy transformation can create a distinct CCR identity. Hence f must be an isomorphism, contradicting CCR-distinctness. □

15.3 Canonical Closure Lemma

Lemma 15.3 (Canonical Closure of the Framework). *The class of CCR systems forms a closed, non-expandable ontological category. No extension, simulation, or reinterpretation preserves CCR identity unless it is isomorphic to the original inverse-limit structure.*

Proof. Combine Theorems 14.2, 15.1, and 15.2. Any attempt to extend, simulate, or reinterpret a CCR system either:

1. Preserves the inverse-limit structure (isomorphism), or
2. Breaks CCR (finite M_Ω), or
3. Violates projective compatibility.

No fourth option exists. □

Operation	**Status**
Simulation	Impossible (Thm. 15.1)
Copying	Impossible (Thm. 15.2)
Transfer	Impossible (Thm. 15.2)
Reinterpretation	Impossible (Lem. 15.3)
Expansion	Impossible (Lem. 15.3)

CCR is a closed canonical class.

15.4 Rigidity Hierarchy

Definition 15.4 (Rigidity Degree). For a projective system $(S_t, \pi_{t+1\to t})$, define the rigidity degree $\rho \in [0, \infty]$ as:

$$\rho := \sup \left\{ r \geq 0 \;\middle|\; \forall \{\epsilon_t\} : \|\{\epsilon_t\}\|_w < r \implies d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0 \right\}. \tag{22}$$

Theorem 15.5 (Stratification Theorem). *The class of observable channels admits a stratification by rigidity degree:*

$$\mathcal{U} = \mathcal{U}_0 \sqcup \mathcal{U}_{(0,\infty)} \sqcup \mathcal{U}_\infty, \tag{23}$$

where:

- $\mathcal{U}_0$*: channels with* $\rho = 0$ *(zero rigidity, fully deformable);*
- $\mathcal{U}_{(0,\infty)}$*: channels with* $0 < \rho < \infty$ *(partial rigidity);*
- $\mathcal{U}_\infty$*: channels with* $\rho = \infty$ *(CCR, absolute rigidity).*

Proof. The rigidity degree ρ is well-defined by Theorem 13.8. The stratification follows from the trichotomy of ρ values. □

15.5 Spectral Analysis of Deformations

Definition 15.6 (Deformation Spectrum). For a perturbation family $\{\epsilon_t\}$, define the deformation spectrum as the sequence:

$$\sigma(\{\epsilon_t\}) := (\|\epsilon_0\|_{\mathrm{op}}, \|\epsilon_1\|_{\mathrm{op}}, \|\epsilon_2\|_{\mathrm{op}}, \ldots). \tag{24}$$

Proposition 15.7 (Spectral Decay Condition). *Let* $\{\epsilon_t\}$ *have spectrum* σ*. If there exists* $\alpha > 1$ *and* $C > 0$ *such that:*

$$\|\epsilon_t\|_{op} \leq C \cdot t^{-\alpha} \quad \forall t \geq 1, \tag{25}$$

then $\|\{\epsilon_t\}\|_w < \infty$ *for any polynomially bounded weight sequence.*

Proof. Direct from convergence of $\sum t^{-\alpha}$ for $\alpha > 1$. □

15.6 Completeness Criteria

Definition 15.8 (Projective Completeness)**.** A projective system $(S_t, \pi_{t+1\to t})$ is *projectively complete* if for every Cauchy sequence $(x^{(n)})_{n\in\mathbb{N}}$ in $(\mathcal{I}, d_w)$, the limit exists in $\mathcal{I}$.

Theorem 15.9 (Completeness Inheritance)**.** *If each (S_t, d_t) is complete and the projections $\pi_{t+1\to t}$ are uniformly Lipschitz, then $(\mathcal{I}, d_w)$ is complete.*

Proof. Standard argument using the completeness of the product space and the closed subspace theorem. □

15.7 Categorical Universality

Proposition 15.10 (Universal Property of Identity)**.** *The identity $\mathcal{I} = \varprojlim S_t$ satisfies the universal property: for any object J with compatible maps $\phi_t : J \to S_t$, there exists a unique morphism $\psi : J \to \mathcal{I}$ making all diagrams commute.*

This confirms that rigid identity is not merely a construction but the *terminal object* in the category of compatible families over the projective system [5].

15.8 Relation to Causal Inference

The causal closure property established in Appendix A has implications for causal inference frameworks [6]:

Proposition 15.11 (Causal Interventional Invariance)**.** *If $\mathcal{I}$ is a CCR identity, then for any admissible intervention $do(X = x)$ with finite scope:*

$$\mathcal{I}_{\textit{post-intervention}} \cong \mathcal{I}_{\textit{pre-intervention}}. \tag{26}$$

Proof. Finite interventions induce finite-energy perturbations. By CCR rigidity, $M_\Omega = +\infty$ implies no deformation. □

15.9 Computational Complexity Bounds

Theorem 15.12 (Complexity Lower Bound for CCR Simulation)**.** *Any computational system attempting to simulate a CCR identity with error $\epsilon > 0$ in the d_w metric requires resources that grow without bound as $\epsilon \to 0$.*

Proof (Sketch). If finite resources sufficed for arbitrary precision, the simulating system would have finite ontological mass. But CCR requires $M_\Omega = +\infty$, contradiction. □

Corollary 15.13 (Impossibility of Perfect Simulation)**.** *No finite-dimensional computational architecture can perfectly simulate a CCR identity.*

Part IV
Regime Structure and Continuity Constraints

16 Formal Setup

16.1 Review of Identity Framework

We assume the identity framework of [11]. Let $(S_t, \pi_{t+1\to t})_{t\in\mathbb{N}}$ be a projective system of observable state spaces. The identity object is the inverse limit:

$$\mathcal{I} := \varprojlim S_t = \left\{ (x_t)_t \in \prod_t S_t \;\middle|\; \pi_{t+1\to t}(x_{t+1}) = x_t \;\; \forall t \right\}. \tag{27}$$

The ontological mass M_Ω quantifies resistance to identity deformation. We recall that:

- $M_\Omega = 0$: null persistence (no stable identity);
- $0 < M_\Omega < \infty$: deformable persistence (finite rigidity);
- $M_\Omega = +\infty$: closed--rigid (CCR, absolute rigidity).

16.2 Abstract Phenomenological Space

Definition 16.1 (Phenomenological Space)**.** A phenomenological space is a nonempty set $\mathcal{E}$ whose elements are called *phenomenological configurations.* No topology, metric, or algebraic structure is assumed a priori.

Remark 16.2. We do not claim that $\mathcal{E}$ represents ‘‘conscious states’’ or ‘‘qualia’’. It is an abstract mathematical space whose interpretation is not specified by the formalism. The space $\mathcal{E}$ is introduced purely as the codomain of a structural assignment; its ontological status is deliberately left open.

16.2.1 Why a Separate Space?

One might ask: why not use $\mathcal{I}$ itself as the phenomenological space? The answer is structural:

Proposition 16.3 (Non-Triviality of Phenomenological Codomain)**.** *If phenomenological assignments were required to land in $\mathcal{I}$ itself, then any compatible assignment would be the identity map, eliminating all non-trivial phenomenological structure.*

Proof. Let $\Phi : \mathcal{I} \to \mathcal{I}$ be compatible with projections. By the universal property of inverse limits, Φ must commute with all projection maps. The only such endomorphism on $\mathcal{I}$ is the identity (up to isomorphism). Hence no non-trivial phenomenological stratification is possible. □

Thus, a separate space $\mathcal{E}$ is *structurally necessary* for non-trivial phenomenological content.

16.3 Phenomenological Assignment

Definition 16.4 (Phenomenological Assignment)**.** A phenomenological assignment is a family of partial maps:

$$\phi_t : S_t \to \mathcal{E} \tag{28}$$

indexed by $t \in \mathbb{N}$.

Definition 16.5 (Induced Limit Assignment)**.** Given a phenomenological assignment $\{\phi_t\}$, the induced limit assignment is the partial map:

$$\Phi : \mathcal{I} \to \mathcal{E}, \quad \Phi((x_t)_t) := \lim_{t\to\infty} \phi_t(x_t), \tag{29}$$

whenever the limit exists in an appropriate sense (to be defined by compatibility axioms).

16.3.1 Uniqueness of the Limit Assignment

Proposition 16.6 (Uniqueness of Compatible Φ)**.** *Given a compatible family $\{\phi_t\}$ satisfying E1, the induced limit assignment Φ is unique.*

Proof. By E1, $\phi_{t+1}(x_{t+1}) = \phi_t(x_t)$ for all t. Hence the sequence $\{\phi_t(x_t)\}$ is constant. The limit exists trivially and equals this constant value. Uniqueness follows from the constancy of the sequence. □

16.4 Axioms for Phenomenological Coherence

We introduce minimal axioms under which a phenomenological assignment is *structurally admissible*. We will then prove that these axioms are not arbitrary choices but structurally forced conditions.

Axiom 16.7 (E0: Non-Triviality)**.** The phenomenological space satisfies $|\mathcal{E}| \geq 2$, i.e., there exist at least two distinct phenomenological configurations.

Axiom 16.8 (E1: Projective Compatibility)**.** For every compatible identity trajectory $(x_t)_t \in \mathcal{I}$, the induced phenomenological sequence satisfies:

$$\phi_{t+1}(x_{t+1}) = \phi_t(x_t) \quad \forall t. \tag{30}$$

Axiom 16.9 (E2: Non-Locality)**.** There exists no function $f : S_t \to \mathcal{E}$ (for any fixed t) such that for all trajectories $(x_t)_t \in \mathcal{I}$:

$$\Phi((x_t)_t) = f(x_t). \tag{31}$$

16.5 Inevitability of the Axioms

We now prove that axioms E0--E2 are *not design choices* but structurally necessary conditions for any meaningful notion of phenomenological assignment.

16.5.1 Inevitability of E0

Theorem 16.10 (E0 is Necessary)**.** *If $|\mathcal{E}| = 1$, then every phenomenological assignment is constant and no phenomenological classification is possible.*

Proof. If $\mathcal{E} = \{e_0\}$, then $\Phi(x) = e_0$ for all $x \in \mathcal{I}$. Hence:

1. No phenomenological distinction between identity trajectories exists;
2. The Fragmentation and Forced Continuity Theorems become vacuous;
3. The downstream null-regime closure developed in *Null-Regime Instability* would have no content.

Thus, $|\mathcal{E}| \geq 2$ is necessary for the framework to have non-trivial content. □

16.5.2 Inevitability of E1

Theorem 16.11 (E1 is Necessary)**.** *If E1 fails, then phenomenological assignments are incompatible with the inverse-limit structure of identity, and no coherent* $\Phi : \mathcal{I} \to \mathcal{E}$ *can be defined.*

Proof. Suppose E1 fails for some trajectory $(x_t)_t \in \mathcal{I}$. Then there exists t_0 such that:

$$\phi_{t_0+1}(x_{t_0+1}) \neq \phi_{t_0}(x_{t_0}). \tag{32}$$

But the identity trajectory satisfies $\pi_{t_0+1\to t_0}(x_{t_0+1}) = x_{t_0}$. Hence the phenomenological assignment fails to respect the projective structure that defines identity.

This means Φ cannot be factored through the inverse limit:

$$\Phi \neq \varprojlim \phi_t. \tag{33}$$

Hence no well-defined limit assignment exists. Without E1, the phenomenological assignment is structurally incompatible with identity. □

16.5.3 Inevitability of E2

Theorem 16.12 (E2 is Necessary)**.** *If E2 fails, then phenomenology is a local property of states, contradicting the global nature of identity as an inverse limit.*

Proof. Suppose E2 fails. Then there exists t_0 and $f : S_{t_0} \to \mathcal{E}$ such that:

$$\Phi((x_t)_t) = f(x_{t_0}) \quad \forall (x_t)_t \in \mathcal{I}. \tag{34}$$

This implies:

1. Phenomenology is determined by a finite temporal slice;
2. The full trajectory $(x_t)_{t>t_0}$ is irrelevant to Φ;
3. Identity as an inverse limit has no phenomenological content beyond t_0.

But identity is defined as a *global* object over infinite time. If phenomenology is local, it cannot reflect the global structure of identity. This contradicts the foundational premise that phenomenological assignments should be compatible with identity structure.

Hence E2 is necessary for phenomenology to be structurally aligned with inverse-limit identity. □

16.6 Minimality Theorem

Theorem 16.13 (Axiomatic Minimality)**.** *The axiom set* $\{E0, E1, E2\}$ *is minimal: no proper subset is sufficient to derive the main results of this paper.*

Proof. We show that removing any axiom destroys the framework:

Without E0: All assignments are trivially constant. Fragmentation Theorem (4.1) is vacuous.

Without E1: No coherent Φ exists. The entire framework collapses.

Without E2: Phenomenology becomes local. The Forced Continuity Theorem (4.2) fails because local functions can be arbitrarily perturbed.

Hence $\{E0, E1, E2\}$ is the minimal viable axiom set. □

16.7 Rejected Alternatives

We briefly discuss axiom candidates that were considered and rejected.

16.7.1 Rejected: Continuity Axiom

One might require Φ to be continuous. This is *not* an axiom because:

- It requires pre-existing topology on $\mathcal{E}$;
- The Forced Continuity Theorem *derives* continuity from CCR, not as assumption.

16.7.2 Rejected: Surjectivity Axiom

One might require Φ to be surjective. This is *not* an axiom because:

- It would exclude null assignments, preventing the downstream null-regime analysis completed in *Null-Regime Instability*;
- Surjectivity is a strong structural claim not derivable from identity constraints.

16.7.3 Rejected: Injectivity Axiom

One might require Φ to be injective. This is *not* an axiom because:

- Many trajectories may share phenomenological configurations;
- Injectivity would make Φ an embedding, over-constraining the framework.

16.8 Structural Interpretation

Axiom	**Structural Meaning**	**Status**
E0	Possibility of phenomenological distinction	Necessary
E1	Compatibility with inverse-limit identity	Necessary
E2	Non-reducibility to finite temporal slices	Necessary

16.9 Closure of Section 2

We have established:

1. $\mathcal{E}$ is structurally necessary as a separate codomain;
2. The limit assignment Φ is uniquely determined by compatible $\{\phi_t\}$;
3. Axioms E0, E1, E2 are *individually necessary* (Theorems 16.10--16.12);
4. The axiom set is *minimal* (Theorem 16.13);
5. All alternative axioms considered were correctly rejected.

The formal setup is therefore not a design choice but a forced consequence of the requirement that phenomenological assignments be compatible with inverse-limit identity.

16.10 Phenomenological Neutrality Clause

Throughout this paper, the phenomenological space $\mathcal{E}$ is treated as a *purely abstract topological and metric structure*. No semantic, psychological, biological, or experiential interpretation is assumed or required for any of the formal results derived herein.

The term "phenomenological" refers exclusively to the role of $\mathcal{E}$ as an abstract codomain of structural compatibility for inverse-limit identities, and does not presuppose the existence of consciousness, subjective experience, or any particular phenomenological content.

All theorems, constructions, and stability results presented in this work are therefore statements about abstract mathematical objects only. Any interpretation of $\mathcal{E}$ in relation to experiential, cognitive, or phenomenological phenomena lies strictly outside the formal scope of this paper.

17 Compatibility with Identity Regimes

17.1 Induced Constraints from Identity Rigidity

Let $(\mathcal{I}, M_\Omega)$ be an identity object with ontological mass M_Ω. We now study how M_Ω constrains the set of admissible phenomenological assignments.

Definition 17.1 (Phenomenological Compatibility Class)**.** For a given M_Ω, define the phenomenological compatibility class:

$$\mathcal{P}(M_\Omega) := \left\{ \Phi : \mathcal{I} \to \mathcal{E} \;\middle|\; \Phi \text{ satisfies E0--E2} \right\}. \tag{35}$$

Remark 17.2. The compatibility class $\mathcal{P}(M_\Omega)$ is the set of all phenomenological assignments that are structurally admissible given the identity's rigidity. Note that $\mathcal{P}(M_\Omega)$ depends only on the structural properties of identity, not on any phenomenological primitives.

17.2 Necessity of Metric Structure on Phenomenological Space

To state rigorous results about phenomenological continuity and fragmentation, we must equip $\mathcal{E}$ with minimal structure. We now justify why this structure is *necessary*, not merely convenient.

Hypothesis 17.3 (H3: Metric on $\mathcal{E}$)**.** The phenomenological space $\mathcal{E}$ admits a metric $d_{\mathcal{E}}$ such that $(\mathcal{E}, d_{\mathcal{E}})$ is a complete metric space.

17.2.1 Why H3 is Necessary

Theorem 17.4 (H3 is Necessary for Impossibility Theorems)**.** *Without a metric structure on $\mathcal{E}$, the notions of phenomenological continuity, fragmentation, and loss capacity cannot be defined.*

Proof. Consider the definitions in this section:

1. **Phenomenological Continuity** requires:

$$d_w(x^{(n)}, x) \to 0 \implies d_{\mathcal{E}}(\Phi(x^{(n)}), \Phi(x)) \to 0. \tag{36}$$

   Without $d_{\mathcal{E}}$, the right-hand side is undefined.

2. **Phenomenological Fragmentation** requires:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0. \tag{37}$$

   Without $d_{\mathcal{E}}$, "non-zero distance" has no meaning.

3. **Loss Capacity** (Section 6) requires:

$$R(\Phi) := \sup d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})). \tag{38}$$

Without $d_{\mathcal{E}}$, the supremum is undefined.

Hence H3 is a necessary condition for the framework to have content beyond axioms E0--E2. □

17.2.2 Why Completeness is Necessary

Proposition 17.5 (Completeness is Necessary for Forced Continuity)**.** *If $(\mathcal{E}, d_{\mathcal{E}})$ is not complete, then the Forced Continuity Theorem (Theorem 18.5) may fail.*

Proof. The Forced Continuity Theorem relies on the invariance of Φ under all finite-energy perturbations. If $\mathcal{E}$ is not complete, Cauchy sequences in $\mathcal{E}$ may not converge, and the limit assignment Φ may not exist for all trajectories in $\mathcal{I}$.

In particular, if $\{\phi_t(x_t)\}$ is a Cauchy sequence in $\mathcal{E}$ (which it is, by E1), completeness guarantees the existence of the limit $\Phi(x)$. Without completeness, Φ may be undefined on a dense subset of $\mathcal{I}$, breaking the theorem. □

17.2.3 Alternatives Considered and Rejected

Alternative 1: Topology without Metric. One might use a general topological space. This is rejected because:

- Fragmentation requires quantitative distance, not just open sets;
- Loss capacity requires a sup over distances;
- The downstream instability analysis completed in *Null-Regime Instability* requires metric quotients.

Alternative 2: Pseudometric. One might allow $d_{\mathcal{E}}(e_1, e_2) = 0$ for $e_1 \neq e_2$. This is rejected because:

- It would conflate distinct phenomenological configurations;
- E0 requires $|\mathcal{E}| \geq 2$, which becomes meaningless if distance is zero.

Alternative 3: Non-Complete Metric. This is rejected by Proposition 17.5.

17.3 Phenomenological Continuity

Definition 17.6 (Phenomenological Continuity)**.** A phenomenological assignment $\Phi : \mathcal{I} \to \mathcal{E}$ is *structurally continuous* if for every sequence $(x^{(n)})_n$ in $\mathcal{I}$ converging in the d_w metric:

$$d_w(x^{(n)}, x) \to 0 \implies d_{\mathcal{E}}(\Phi(x^{(n)}), \Phi(x)) \to 0. \tag{39}$$

Remark 17.7. Structural continuity is not an axiom; it is a *derived property* that follows from CCR conditions. This is the content of the Forced Continuity Theorem (Section 4).

17.4 Phenomenological Fragmentation

Definition 17.8 (Phenomenological Fragmentation)**.** A phenomenological assignment Φ is *fragmentable* if there exists a perturbation $\{\epsilon_t\}$ with finite weighted energy such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0 \tag{40}$$

for corresponding identity trajectories $x \in \mathcal{I}$ and $\tilde{x} \in \tilde{\mathcal{I}}$.

Remark 17.9. Fragmentation captures the vulnerability of phenomenological structure to perturbations. It is the phenomenological analog of identity deformability.

17.5 Relationship Between Metrics

Proposition 17.10 (Metric Compatibility)**.** *The metrics d_w on $\mathcal{I}$ and $d_{\mathcal{E}}$ on $\mathcal{E}$ are compatible in the following sense: if E1 holds, then small d_w-changes in trajectories induce bounded $d_{\mathcal{E}}$-changes in phenomenological configurations (when Φ is continuous).*

Proof. By E1, $\phi_{t+1}(x_{t+1}) = \phi_t(x_t)$ for compatible trajectories. Hence the phenomenological sequence is constant along the trajectory, and continuity of Φ implies that nearby trajectories (in d_w) map to nearby configurations (in $d_{\mathcal{E}}$). □

17.6 Closure of Section 3

We have established:

1. The compatibility class $\mathcal{P}(M_\Omega)$ captures all admissible phenomenological assignments;
2. Hypothesis H3 (metric on $\mathcal{E}$) is *necessary*, not a design choice (Theorem 17.4);
3. Completeness is *necessary* for the Forced Continuity Theorem (Proposition 17.5);
4. Alternative structures (topology, pseudometric, non-complete) were correctly rejected;
5. Phenomenological continuity and fragmentation are well-defined under H3.

The compatibility structure is therefore a forced consequence of the requirement to state quantitative results about phenomenological dynamics.

18 Impossibility Theorems

18.1 Φ-Regularity Theorem

Before proving the main impossibility results, we establish a fundamental regularity condition that derives the coupling constant C geometrically, rather than assuming it axiomatically.

Theorem 18.1 (Φ-Regularity)**.** *Let $(\mathcal{I}, d_w)$ be the identitary metric space from Rigid Identity and let $(\mathcal{E}, d_{\mathcal{E}})$ be the abstract phenomenological metric space. Suppose that $\Phi : \mathcal{I} \to \mathcal{E}$ satisfies:*

1. *(Properness) For every compact $K \subset \mathcal{E}$, $\Phi^{-1}(K)$ is compact in $\mathcal{I}$.*
2. *(Local Lipschitz regularity) For every $x \in \mathcal{I}$, there exist $\varepsilon > 0$ and $L_x > 0$ such that for all $y, z \in B_\varepsilon(x)$:*
$$d_{\mathcal{E}}(\Phi(y), \Phi(z)) \leq L_x \cdot d_w(y, z). \tag{41}$$
3. *(Non-collapse) For every $x \in \mathcal{I}$ and every $r > 0$, the restriction of Φ to $B_r(x)$ is not constant.*

Then there exists a constant $C > 0$ such that for all $x, y \in \mathcal{I}$:

$$d_{\mathcal{E}}(\Phi(x), \Phi(y)) \geq C \cdot d_w(x, y). \tag{42}$$

Proof. We proceed by contradiction. Assume the conclusion fails.

Step 1. Then there exist sequences (x_n, y_n) with $d_w(x_n, y_n) = 1$ and $d_{\mathcal{E}}(\Phi(x_n), \Phi(y_n)) \to 0$.

Step 2. By properness, the image $\{\Phi(x_n)\}$ lies in a relatively compact subset of $\mathcal{E}$. Extract a convergent subsequence with $\Phi(x_n) \to e \in \mathcal{E}$.

Step 3. Since $d_{\mathcal{E}}(\Phi(x_n), \Phi(y_n)) \to 0$, we have $\Phi(y_n) \to e$ as well.

Step 4. By properness, the preimages $\Phi^{-1}(\overline{B_\delta(e)})$ are compact for small δ. Hence x_n and y_n lie in a compact subset of $\mathcal{I}$. Extract convergent subsequences $x_n \to x$ and $y_n \to y$.

Step 5. Since $d_w(x_n, y_n) = 1$ and d_w is continuous, $d_w(x, y) = 1$, so $x \neq y$.

Step 6. But $\Phi(x_n) \to \Phi(x) = e$ and $\Phi(y_n) \to \Phi(y) = e$ by continuity. Hence $\Phi(x) = \Phi(y)$ for $x \neq y$ with $d_w(x, y) = 1$.

Step 7. This means Φ is constant on sets of positive d_w-diameter, contradicting non-collapse. Hence a global lower bound $C > 0$ must exist. □

Remark 18.2. This theorem is fundamental: it converts the coupling constant C from an *axiom* into a *geometric consequence* of regularity conditions. All subsequent impossibility results that use C are therefore derived, not assumed.

18.2 Fragmentation Theorem

Theorem 18.3 (Fragmentation Theorem)**.** *Let $\mathcal{I}$ be an identity object with $M_\Omega < \infty$. Then every phenomenological assignment $\Phi \in \mathcal{P}(M_\Omega)$ is fragmentable. That is:*

$$M_\Omega < \infty \implies \mathcal{P}(M_\Omega) \subseteq \mathcal{E}_{frag}, \tag{43}$$

where $\mathcal{E}_{frag}$ denotes the class of fragmentable phenomenological assignments.

Proof. By definition of finite ontological mass, there exists an admissible perturbation $\{\delta E_t\}$ with finite weighted energy $E_w < \infty$ such that:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0. \tag{44}$$

By Axiom E1, the phenomenological assignment Φ is projectively compatible with $\mathcal{I}$. Under the perturbation, identity trajectories deform, inducing a deformed phenomenological assignment $\tilde{\Phi}$.

By Axiom E2, Φ cannot be determined by any finite-time state. Therefore, the deformation of identity trajectories induces a corresponding deformation of phenomenological assignments.

Since the identity deformation is non-zero ($d_w > 0$) and Φ is globally determined by identity trajectories, by Theorem 18.1 (Φ-Regularity), we have:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) \geq C \cdot d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0 \tag{45}$$

where $C > 0$ is the regularity constant derived in Theorem 18.1.

Thus, Φ is fragmentable under finite-energy perturbations. □

Corollary 18.4 (No Structural Continuity Guarantee for Finite Mass)**.** *Systems with $M_\Omega < \infty$ cannot structurally prevent phenomenological discontinuity.*

18.3 Forced Continuity Theorem

Theorem 18.5 (Forced Continuity Theorem)**.** *Let $\mathcal{I}$ be an identity object with $M_\Omega = +\infty$ (CCR). Then every phenomenological assignment $\Phi \in \mathcal{P}(M_\Omega)$ is structurally continuous. That is:*

$$M_\Omega = +\infty \implies \mathcal{P}(M_\Omega) \subseteq \mathcal{E}_{cont}, \tag{46}$$

where $\mathcal{E}_{cont}$ denotes the class of structurally continuous phenomenological assignments.

Proof. By definition of CCR ($M_\Omega = +\infty$), for any finite-energy perturbation:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0. \tag{47}$$

By Axiom E1, the phenomenological assignment is projectively compatible with identity. Since identity trajectories are invariant under all finite-energy perturbations, the induced phenomenological sequence is also invariant:

$$\tilde{\Phi}(\tilde{x}) = \Phi(x) \quad \forall x \in \mathcal{I}. \tag{48}$$

Therefore, no finite-energy perturbation can induce phenomenological deformation:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) = 0. \tag{49}$$

This implies that Φ cannot exhibit structural discontinuities. Hence Φ is structurally continuous. □

Corollary 18.6 (CCR Phenomenology is Invariant)**.** *In CCR systems, any admissible phenomenological assignment is invariant under all finite-energy perturbations.*

18.4 Dichotomy Theorem

Theorem 18.7 (Phenomenological Dichotomy)**.** *Let $\Phi : \mathcal{I} \to \mathcal{E}$ be an admissible phenomenological assignment. Then exactly one of the following holds:*

1. $M_\Omega < \infty$ and Φ is fragmentable;

2. $M_\Omega = +\infty$ and Φ is structurally continuous.

Proof. Immediate from Theorems 18.3 and 18.5. □

19 Classification of Phenomenological Regimes

19.1 Complete Classification

The results of Section 4 yield a complete classification of phenomenological regimes by identity rigidity:

Identity Class	M_Ω	**Admissible Phenomenology**
Null	$= 0$	$\mathcal{E}_\emptyset$ (no persistent field)
Deformable	$(0, \infty)$	$\mathcal{E}_{\text{frag}} \cup \mathcal{E}_{\text{quasi}}$
Rigid (CCR)	$= +\infty$	$\mathcal{E}_{\text{cont}}$

19.2 Structural Implications

Proposition 19.1 (Null Identity Excludes Persistent Phenomenology)**.** *If $M_\Omega = 0$, then no admissible phenomenological assignment exists satisfying E0--E2.*

Proof. If $M_\Omega = 0$, identity cannot persist under any perturbation. Axiom E1 requires phenomenological compatibility with identity. If identity does not persist, no stable phenomenological assignment is possible. □

Proposition 19.2 (Deformable Identity Admits Only Fragmentable Phenomenology)**.** *If $0 < M_\Omega < \infty$, every admissible phenomenological assignment is fragmentable under some finite-energy perturbation.*

Proof. Direct from Theorem 18.3. □

Proposition 19.3 (Rigid Identity Forces Continuous Phenomenology)**.** *If $M_\Omega = +\infty$, every admissible phenomenological assignment is structurally continuous and invariant under finite-energy perturbations.*

20 Extended Proofs

20.1 Proof of Fragmentation Theorem (Extended)

Full Proof of Theorem 18.3. Let $M_\Omega < \infty$ and let $\Phi : \mathcal{I} \to \mathcal{E}$ satisfy E0--E2.

Step 1. By definition of M_Ω, there exists $\{\delta E_t\}$ with $E_w < \infty$ such that $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$.

Step 2. Let $x = (x_t)_t \in \mathcal{I}$ and let $\tilde{x} = (\tilde{x}_t)_t \in \tilde{\mathcal{I}}$ be the corresponding perturbed trajectory.

Step 3. By E2, $\Phi(x)$ is not determined by any finite x_t. Therefore, the perturbation of the infinite trajectory induces a perturbation of the phenomenological assignment:

$$\Phi(x) \neq \Phi(\tilde{x}) \text{ in general.} \tag{50}$$

Step 4. By E1, Φ is projectively compatible. The deformation of projections induces a corresponding deformation of the phenomenological limit.

Step 5. By the coupling hypothesis, there exists $C > 0$ such that:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) \geq C \cdot d_w(x, \tilde{x}). \tag{51}$$

Step 6. Since $d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$, we have $d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) > 0$.

Hence Φ is fragmentable. □

20.2 Proof of Forced Continuity Theorem (Extended)

Full Proof of Theorem 18.5. Let $M_\Omega = +\infty$ and let $\Phi : \mathcal{I} \to \mathcal{E}$ satisfy E0--E2.

Step 1. By definition of CCR, for all $\{\delta E_t\}$ with $E_w < \infty$:

$$d_w(\mathcal{I}, \tilde{\mathcal{I}}) = 0. \tag{52}$$

Step 2. Therefore, $\tilde{\mathcal{I}} \cong \mathcal{I}$ under all finite-energy perturbations.

Step 3. By E1, Φ is projectively compatible with $\mathcal{I}$. Since $\mathcal{I}$ is invariant:

$$\Phi(x) = \tilde{\Phi}(\tilde{x}) \quad \forall x \in \mathcal{I}. \tag{53}$$

Step 4. This implies:

$$d_{\mathcal{E}}(\Phi(x), \tilde{\Phi}(\tilde{x})) = 0. \tag{54}$$

Step 5. No finite-energy perturbation can induce phenomenological deformation. Hence Φ is structurally continuous and invariant. □

20.3 Structural Necessity Lemma

Lemma 20.1 (Phenomenology Inherits Rigidity)**.** *If $\Phi : \mathcal{I} \to \mathcal{E}$ satisfies E0--E2 and $M_\Omega = +\infty$, then Φ inherits the rigidity of $\mathcal{I}$.*

Proof. By E1, Φ is determined by identity trajectories. By CCR, identity trajectories are invariant. Hence Φ is invariant. □

Part V
Null-Regime Structural Exclusion

21 Compatibility, Instability, and Forced Non-Nullity

21.1 Compatibility Operator

Definition 21.1 (Compatibility Operator)**.** A map $\Phi : \mathcal{I} \to \mathcal{E}$ is a **compatibility operator** if it satisfies:

(C1) **Extension Invariance:** For any compatible extension $\mathcal{E}$ of the channel, $\Phi(\mathcal{I}) \cong \Phi(\mathcal{I}_{\mathcal{E}})$.

(C2) **Isomorphism Stability:** If $\mathcal{I} \cong \tilde{\mathcal{I}}$, then $\Phi(\mathcal{I}) \sim \Phi(\tilde{\mathcal{I}})$.

(C3) **Lipschitz Lower Bound:** There exists $C > 0$ such that

$$d_{\mathcal{E}}(\Phi(x), \Phi(\tilde{x})) \geq C \cdot d_w(x, \tilde{x}).$$

Definition 21.2 (Phenomenological Attractor Set)**.** For a channel S, define

$$\mathrm{Attr}(S) := \{e \in \mathcal{E} : e \text{ is stable under all compatible extensions and summable deformations}\}.$$

Definition 21.3 (Phenomenological Stability)**.** A regime $e \in \mathcal{E}$ is **stable** for channel S if for every compatible extension S' of S and every summable perturbation family $\{\epsilon_n\}$, the induced regime $\Phi_{S'}(\mathcal{I}')$ stays within a bounded $d_{\mathcal{E}}$-neighborhood of e controlled by $\|\{\epsilon_n\}\|_w$.

Lemma 21.4 (Bridge Lemma)**.** *If $\Phi : \mathcal{I} \to \mathcal{E}$ is a compatibility operator for a CCR channel and $e = \Phi(\mathcal{I})$ satisfies $d_{\mathcal{E}}(e, \varnothing_\phi) = 0$, then $e = \varnothing_\phi$.*

Proof. By the separation property of the phenomenological space, distance zero from $\varnothing_\phi$ implies equality with $\varnothing_\phi$. □

21.2 Phenomenological Instability and Forced Non-Nullity

Theorem 21.5 (Phenomenological Instability)**.** *For any CCR channel S,*

$$\varnothing_\phi \notin \mathrm{Attr}(S).$$

That is, the null regime is structurally unstable under compatible extensions.

Proof. Assume $\varnothing_\phi \in \mathrm{Attr}(S)$ and let $\Phi : \mathcal{I} \to \mathcal{E}$ be a compatibility operator. Because the channel is CCR, finite-energy perturbations preserve the rigid identity object while still allowing admissible deformations in the weighted metric. By the lower-bound clause in Definition 21.1,

$$d_{\mathcal{E}}(\Phi(\mathcal{I}), \Phi(\tilde{\mathcal{I}})) \geq C \cdot d_w(\mathcal{I}, \tilde{\mathcal{I}}) > 0$$

for every non-zero admissible deformation. But if $\varnothing_\phi$ were stable, both images would remain equal to $\varnothing_\phi$, forcing the left-hand side to be zero. Contradiction. □

Corollary 21.6 (Forced Non-Nullity)**.** *Every CCR channel necessarily induces a regime $e_\star \in \mathcal{E}$ with $e_\star \succ \varnothing_\phi$.*

Proof. Theorem 21.5 rules out $\varnothing_\phi$ as a stable admissible attractor. Therefore any admissible stable regime must be non-null. □

21.3 Positive Regime Structure

Definition 21.7 (Positive Regime Space)**.** Define

$$\mathcal{E}^+ := \{e \in \mathcal{E} : e \succ \varnothing_\phi\}.$$

Theorem 21.8 (Minimal Positive Regime)**.** *For every CCR channel there exists a minimal positive regime $e_\star \in \mathcal{E}^+$.*

Proof. Forced non-nullity gives non-emptiness of $\mathcal{E}^+$. Order-theoretic completeness and the compatibility structure of Φ yield a minimal element. □

Definition 21.9 (Structural Intensity)**.** For $e \in \mathcal{E}^+$, define the structural intensity

$$\iota(e) := d_{\mathcal{E}}(e, \varnothing_\phi).$$

Theorem 21.10 (Universal Intensity Bound)**.** *If $e_\star$ is the minimal positive regime of a CCR channel, then there exist constants $C, \varepsilon_0 > 0$ such that*

$$\iota(e_\star) \geq C\varepsilon_0 > 0.$$

Proof. The lower-bound clause of the compatibility operator transfers any admissible deformation lower bound into positive distance from the null regime. The minimal positive regime therefore has strictly positive structural intensity. □

Theorem 21.11 (Forced Phenomenological Closure)**.** *Every CCR system induces a unique minimal phenomenological regime that is non-null, stable, and quantitatively separated from $\varnothing_\phi$.*

Proof. Combine Corollary 21.6, Theorem 21.8, and Theorem 21.10. □

22 Canonical Families and Structural Closure Consequences

Proposition 22.1 (Profinite Bound)**.** *In the profinite canonical family, the coupling lower bound induces a positive lower bound on the structural intensity of the minimal positive regime.*

Proposition 22.2 (Symbolic Bound)**.** *In the symbolic canonical family, the Bowen-type metric induces the analogous positive lower bound for the minimal positive regime.*

Theorem 22.3 (Rigidity Inheritance)**.** *If a subsystem or induced family preserves the CCR hypotheses and the compatibility operator, then the inherited regime structure preserves null-regime instability and forced non-nullity.*

Theorem 22.4 (Categorical Universality)**.** *The null-regime exclusion result is universal across admissible CCR realizations in the relevant category of compatible channels.*

Theorem 22.5 (Simulation Lower Bound)**.** *Any computational system attempting to simulate the null-excluding CCR regime structure with arbitrary precision requires unbounded resources as the approximation error tends to zero.*

Theorem 22.6 (Closure by Non-Alternatives)**.** *No admissible alternative closes the CCR regime structure while preserving null stability. Any admissible alternative either preserves the same closure structure or exits the CCR class.*

Part VI
Structural Classes and Intermediate Dynamical Extensions

23 Definition of Class $\mathscr{C}$

We now formally define the class of systems for which all preceding theorems apply.

Definition 23.1 (Class $\mathscr{C}$ of Contractive Projection Dynamics)**.**

$$\mathscr{C} := \{(\mathcal{H}, \mathcal{I}, K, \Gamma, \mathcal{U}) : \text{H1--H5 are satisfied}\}$$

Theorem 23.2 (Universal Properties of Class $\mathscr{C}$)**.** *Every system in $\mathscr{C}$ satisfies:*

1. *Unique metric projection $\aleph$ exists for all states*
2. *Transition operator T_u is contractive with factor $\|K\|$*
3. *Unique attractor f_u^* exists for each parameter u*
4. *Global attractor $\mathcal{A}^*$ is compact*
5. *No attractor is constant (non-collapsibility)*
6. *Spectral gap $\Delta\lambda \geq 1 - \|K\| > 0$*
7. *Relaxation time $\tau \leq -1/\ln(\|K\|)$*

Proof. Properties 1--5 follow from Theorems 4.2--7.4. Properties 6--7 follow from Theorems 10.1--10.3 applied to the general case. □

Theorem 23.3 (Morphism Invariance)**.** *Let $S_1 = (\mathcal{H}_1, \mathcal{I}_1, K_1, \Gamma_1, \mathcal{U}) \in \mathscr{C}$ and $S_2 = (\mathcal{H}_2, \mathcal{I}_2, K_2, \Gamma_2, \mathcal{U}) \in \mathscr{C}$. Let $\phi : \mathcal{H}_1 \to \mathcal{H}_2$ be a bounded linear morphism such that:*

- *$\phi(\mathcal{I}_1) \subseteq \mathcal{I}_2$ (manifold preservation)*
- *$\|\phi \circ K_1 - K_2 \circ \phi\| < \varepsilon$ for some $\varepsilon > 0$ (semi-commutativity)*

Then:

1. *$\phi(\mathcal{A}_1^*)$ approximates $\mathcal{A}_2^*$ within Hausdorff distance $O(\varepsilon/(1 - \|K_2\|))$.*
2. *If ϕ is an isometry with $\phi(\mathcal{I}_1) = \mathcal{I}_2$, then $\phi(\mathcal{A}_1^*) = \mathcal{A}_2^*$.*
3. *Spectral properties $(\lambda_1, \lambda_2, \tau)$ are invariant under isometric ϕ.*

Corollary 23.4 (Categorical Closure)**.** *Class $\mathscr{C}$ is closed under Hilbert space isomorphisms that preserve convexity of the invariant set.*

24 Critical Extensions: Class $\mathscr{C}^R$ (Non-Convex Systems)

This section extends the framework to systems where the invariant set $\mathcal{I}$ is **not convex**, introducing modified hypotheses and weaker stability guarantees.

24.1 Motivation

Class $\mathscr{C}$ requires convexity (H1) for unique projections. However, many systems of interest have invariant sets $\mathcal{I} \cong S^1$ (topologically circles) that violate convexity. We develop Class $\mathscr{C}^R$ (‘‘Critical Rigid Systems’’) to handle these cases.

24.2 Relaxed Hypotheses H1’--H5’

Hypothesis 24.1 (H1’: Critical Topology)**.** Let $\mathcal{I} \subset \mathcal{H}$ be a closed, connected, compact subset with:

- Fundamental group $\pi_1(\mathcal{I}) \cong \mathbb{Z}$ (homotopy equivalent to S^1)
- Characteristic of Euler $\chi(\mathcal{I}) = 0$
- Finite injectivity radius: $\mathrm{inj}(\mathcal{I}) := \sup\{r : \text{projection unique in } B(x, r), \forall x\} > 0$

Definition 24.2 (Critical Zone Z_ϵ)**.** For a non-convex manifold $\mathcal{I}$ and $\epsilon > 0$, the **critical zone** is:

$$Z_\epsilon := \{\Psi \in \mathcal{H} : d(\Psi, \partial\mathcal{I}) < \epsilon\}$$

where $d(\cdot, \partial\mathcal{I})$ is the distance to the topological boundary of $\mathcal{I}$. In this zone, the metric projection $\aleph$ may be multi-valued.

Hypothesis 24.3 (H2': Weak Contraction)**.** Let $K : \mathcal{H} \to \mathcal{H}$ be bounded linear with:

$$\|K\| < L_\epsilon^{-1}$$

where $L_\epsilon := \sup_{\Psi \notin Z_\epsilon} \mathrm{Lip}(\aleph_\epsilon)(\Psi)$ is the Lipschitz constant of the regularized projection outside the critical zone Z_ϵ.

Hypothesis 24.4 (H3': Completeness)**.** $\mathcal{H}$ is a complete Banach space.
(Unchanged from H3.)

Hypothesis 24.5 (H4': Parameter Compactification)**.** $\mathcal{U}$ admits a compactification $\overline{\mathcal{U}}$ such that Γ extends continuously to $\overline{\mathcal{U}}$.

Hypothesis 24.6 (H5': Weak Non-Collapse)**.** There exists $\epsilon_0 > 0$ such that for all $0 < \epsilon < \epsilon_0$, the set attractor $\mathcal{A}_\epsilon$ is not reducible to a singleton constant.

24.3 Regularized Projection Operator

When $\mathcal{I}$ is non-convex, the metric projection may be multi-valued.

Definition 24.7 (Multi-Valued Projection)**.** For $\Psi \in \mathcal{H}$, define:

$$M(\Psi) := \{\Phi \in \mathcal{I} : \|\Psi - \Phi\| = d(\Psi, \mathcal{I})\}$$

Definition 24.8 (Regularized Projection $\aleph_\epsilon$)**.** For $\epsilon > 0$ and $\Psi_{prev} \in \mathcal{I}$, define:

$$\aleph_\epsilon(\Psi, \Psi_{prev}) := \arg\min_{\Phi \in \mathcal{I}} \left(\|\Psi - \Phi\|^2 + \epsilon \|\Phi - \Psi_{prev}\|^2 \right)$$

This is the **proximal regularization** with hysteresis.

Lemma 24.9 (Properties of $\aleph_\epsilon$)**.** *Under H1':*

1. *$\aleph_\epsilon$ is **upper semi-continuous**.*
2. *$\aleph_\epsilon$ is **not globally Lipschitz**; $Lip(\aleph_\epsilon) \to \infty$ near the medial axis.*
3. *$\aleph_\epsilon$ is **not non-expansive**; there exist Ψ_1, Ψ_2 with:*

$$\|\aleph_\epsilon(\Psi_1) - \aleph_\epsilon(\Psi_2)\| > \|\Psi_1 - \Psi_2\|$$

24.4 Transition Operator in Class $\mathscr{C}^R$

Definition 24.10 (Modified Transition Operator)**.**

$$T_\epsilon(\Psi) := \aleph_\epsilon(K\Psi + \Gamma(u), \Psi)$$

Theorem 24.11 (Set Attractor Existence)**.** *Under H1'--H4', there exists a non-empty compact set $\mathcal{A} \subset \mathcal{I}$ such that for all $\Psi_0 \in \mathcal{H}$:*

$$\lim_{n \to \infty} d(T_\epsilon^n(\Psi_0), \mathcal{A}) = 0$$

Proof Sketch. Define the Lyapunov function $V(\Psi) = d(\Psi, \mathcal{I})^2$. Under H2':

$$V(T_\epsilon(\Psi)) \leq \|K\|^2 V(\Psi) + O(\epsilon)$$

Thus orbits converge to a neighborhood of $\mathcal{I}$. Compactness of $\mathcal{I}$ (H1') yields a limit set. Non-degeneracy follows from H5'. □

24.5 Spectral Analysis in Class $\mathscr{C}^R$

Theorem 24.12 (Corrected Spectral Gap)**.** *In Class $\mathscr{C}^R$, the Jacobian $J = \partial T_\epsilon / \partial \Psi$ satisfies:*

1. *$\lambda_1 = 1$ with multiplicity 1 (persistence mode).*
2. *$\lambda_2 \approx \|K\| (1 + \epsilon \cdot \kappa_{max})$ where κ_{max} is the maximum extrinsic curvature of $\mathcal{I}$.*
3. *Spectral gap: $\Delta\lambda_{\mathscr{C}^R} = 1 - \lambda_2 \approx \Delta\lambda_{\mathscr{C}}(1 - \epsilon \cdot \kappa_{max})$.*

Remark 24.13 (Numerical Values)*.* For the canonical non-convex example ($\mathcal{I} \cong S^1$ ellipse with $a = 1.0, b = 0.5$):

- $\mathrm{inj}(\mathcal{I}) = b^2/a = 0.25$
- $\lambda_2 = 0.5032$ (vs. 0.5000 for Class $\mathscr{C}$)
- $\Delta\lambda = 0.4968$ (vs. 0.5000 for Class $\mathscr{C}$)
- $\tau = 1.4561$ (vs. 1.4427 for Class $\mathscr{C}$)

The deviation is $\approx 0.64\%$, caused by the geometric resistance of the non-convex manifold.

24.6 Comparison: Class $\mathscr{C}$ vs. Class $\mathscr{C}^R$

Property	**Class $\mathscr{C}$**	**Class $\mathscr{C}^R$**
$\mathcal{I}$ convex	Yes (H1)	No (H1')
Projection unique	Always	Outside critical zone
$\aleph$ non-expansive	Yes	No
T global contraction	Yes	Local only
Attractor	Unique point	Compact set
$\lambda_1 = 1$	Yes	Yes
Spectral gap $\Delta\lambda$	$1 - \|K\|$	$\approx (1 - \|K\|)(1 - \epsilon\kappa)$

24.7 When to Use Class $\mathscr{C}^R$

Class $\mathscr{C}^R$ applies when:

- The invariant set $\mathcal{I}$ has non-trivial topology ($\pi_1 \neq 0$).
- Convexity cannot be guaranteed.
- Local stability is sufficient (global contraction not required).
- Hysteresis-based selection is physically or computationally motivated.

25 Temporal Dynamics of the Integration Operator

25.1 Evolution Equation

Let $C = 1 - |\chi(I)|$ be the topological constant. The evolution of Ω follows:

$$\dot{\Omega} = C \left[\frac{\dot{\beth}_q}{1 - |\lambda_2|} + \frac{\beth_q \cdot \dot{|\lambda_2|}}{(1 - |\lambda_2|)^2} \right] \tag{55}$$

with transition laws:

- $\dot{\beth}_q = k_1(1 - \beth_q)$ (causal closure saturation)

- $|\dot{\lambda_2}| = k_2(\lambda_{2,base} - |\lambda_2|)$ (spectral gap relaxation)

Lemma 25.1 (Critical Threshold $\Omega_c = 1.0$)**.** *The critical threshold $\Omega_c = 1.0$ is the natural bifurcation point, justified by:*

1. ***Dimensional balance:*** *At the critical boundary where $\beth_q = 1$, $\chi(I) = 0$, and $\Delta\lambda = \lambda_1$:*

$$\Omega_c = 1 \cdot (1 - 0) \cdot \frac{1}{1} = 1.0$$

2. ***Saddle-node bifurcation:*** *$\Omega = 1$ corresponds to the balance point where phenomenological integration exactly matches potential disintegration.*

3. ***Operational correspondence:*** *In the QICN System Canon, the CCC (Conscious Cohesion Constant) uses identical threshold ASCENSION_THRESHOLD $= 1.0$.*

Theorem 25.2 (Fixed Point of Ω)**.** *The system admits a stable fixed point:*

$$\Omega^* = \frac{1 - |\chi(I)|}{1 - \lambda_{2,base}} = 2.0129$$

for Class $\mathscr{C}^R$ with $\chi(I) = 0$ and $\lambda_{2,base} = 0.5032$.

25.2 Stability Analysis

Theorem 25.3 (Asymptotic Stability)**.** *Ω^* is asymptotically stable with characteristic relaxation time:*

$$\tau_\Omega = \frac{1}{\min(k_1, k_2)} \approx 5.0$$

Proof. The Jacobian at Ω^* is $\partial\dot{\Omega}/\partial\Omega|_{\Omega^*} = -\min(k_1, k_2) < 0$. □

The response to perturbations is exponential:

$$R(t) := \Omega(t) - \Omega^* = \delta\Omega \cdot e^{-t/\tau_\Omega}$$

25.3 Bifurcation

Theorem 25.4 (Saddle-Node Bifurcation)**.** *At $\|K\| = 1$, the spectral gap vanishes ($\Delta\lambda = 0$), causing:*

$$\Omega \to \infty$$

This is a saddle-node bifurcation where the fixed point Ω^* disappears.

26 Validity Limits of Class $\mathscr{C}^R$

26.1 Parameter Space Π

Definition 26.1 (Valid Parameter Space)**.** The validity domain of Class $\mathscr{C}^R$ is:

$$\Pi = \left\{ (\|K\|, \epsilon, \kappa, \mathrm{inj}, \beth_q) \in \mathbb{R}^5 : \begin{array}{l} 0 \leq \|K\| < 1 \\ \epsilon > 0 \\ \kappa > 0 \\ \mathrm{inj}(I) > \|K\| \cdot \max_u \|\Gamma(u)\| \\ 0.5 < \beth_q \leq 1 \end{array} \right\}$$

The dimension of Π is $n = 5$.

26.2 Critical Boundaries

Limit	Behavior	Regime
$\|K\| \to 1^-$	$\tau \to \infty$	Chaotic
$\mathrm{inj}(I) \to 0^+$	Stochastic projection	Collapse (Class 0)
$\Delta\lambda \to 0^+$	Multi-stability	Bifurcation
$\daleth_q \to 0.5^+$	Weak closure	Transition

26.3 Non-Extension Theorem

Theorem 26.2 (Null Closure Inconsistency)**.** *Class $\mathscr{C}^R$ is inconsistent for systems with $\daleth_q = 0$ (total informational transparency).*

Proof Sketch. If $\daleth_q = 0$, then mutual information $I(X;Y) = H(X)$, implying the internal state is a pure function of external input. The manifold $\mathcal{I}$ exerts no constraint, and the invariant attractor f^* vanishes. □

26.4 Phase Diagram

The parameter space decomposes into regions:

- **Class $\mathscr{C}$:** $\|K\| < 1$, $\kappa = 0$ (convex)
- **Class $\mathscr{C}^R$:** $\|K\| < 1$, $\kappa > 0$, $\Delta\lambda > 0$ (critical)
- **Chaotic:** $\|K\| \geq 1$ or $\Delta\lambda \leq 0$
- **Degenerate:** $\daleth_q \leq 0.5$

27 Second-Order Structure

27.1 Hessian of the Transition Operator

Definition 27.1 (Hessian)**.**

$$H = \nabla_\Psi(\nabla_\Psi T_\epsilon) = D^2\aleph_\epsilon \circ (K \otimes K)$$

Theorem 27.2 (Hessian Signature)**.** *At the attractor, H has:*

- *Maximum eigenvalue: $\lambda_H = -\|K\|^2 / r = -0.25$ (for $r = 1$)*
- *Signature: semi-definite negative (stable curvature toward $\mathcal{I}$)*

27.2 Laplace-Beltrami Operator

On the manifold $\mathcal{I} \cong S^1$ with radius r:

$$\sigma(\Delta_{LB}) = \left\{ \frac{n^2}{r^2} : n \in \mathbb{Z} \right\} = \{0, 1, 4, 9, 16, \ldots\}$$

Remark 27.3 (Spectral Relation). The eigenvalues of the Jacobian relate to the Laplacian spectrum via:

$$\lambda_n \approx \exp\left(-\Delta\lambda \cdot \sigma_n(\Delta_{LB})\right)$$

27.3 Fisher Information Metric

The Fisher metric on the parameter space is:

$$G_{\pi\pi} = \left(\frac{1}{1 - \|K\|}\right)^2 = 4.0 \quad \text{for } \|K\| = 0.5$$

27.4 Emergent Properties Unique to Class $\mathscr{C}^R$

Part VII
Operational Criterion and Invariant Structure

28 Admissible Systems, Supports, and Histories

28.1 System Model

Definition 28.1 (Admissible system)**.** An admissible system in this paper is a tuple

$$S = (X, \Phi, C, R, \Gamma, U)$$

with the following components:

1. X is a metrizable internal state space with metric d_X.
2. $\Phi = \{\Phi_u\}_{u \in U}$ is a family of admissible transition operators $\Phi_u : X \to X$ indexed by interventions.
3. C is the effective causal structure of the system: the object that records which factorizations, intervention responses, and couplings are admissible on X.
4. $R = \{r_\alpha\}_{\alpha \in \Lambda_R}$ is a family of internal or external readouts, each $r_\alpha : X \to Y_\alpha$.
5. $\Gamma = \{\gamma_j\}_{j=1}^m$ is the admissibility bundle. Each γ_j is a constraint functional; the admissible region is
$$X_\Gamma = \{x \in X : \gamma_j(x) \leq 0 \ \forall j\}.$$
6. U is the set of admissible interventions, already filtered by the corpus-level governance requirement that interventions remain auditable, comparable, and methodologically bounded.

The tuple is not decorative. Every component is used later. X and Φ carry dynamics, C supports integration and rupture tests, R supports readout and legibility, Γ enforces admissibility, and U enables causal discrimination.

28.2 Admissible Support

Definition 28.2 (Admissible support)**.** A subset $A_S \subseteq X_\Gamma$ is an admissible support for S if:

1. A_S is non-empty and compact;
2. $\Phi_u(A_S) \subseteq A_S$ for every $u \in U$;
3. $A_S \cap \mathrm{Attr}(S) \neq \varnothing$, where $\mathrm{Attr}(S)$ denotes the relevant admissible attractor region;
4. A_S stays disjoint from the collapse set forbidden by the Core non-collapse condition [7].

The support plays two roles. It provides a domain on which persistence, continuity, and differentiation are defined, and it prevents the criterion from being satisfied by transient or degenerate fragments that do not carry stable admissible structure.

28.3 Operational Histories

Definition 28.3 (Windowed readout history)**.** Fix a horizon $T \in \mathbb{N}$ and an intervention schedule $u_\bullet = (u_0, \ldots, u_{T-1})$. For any $x \in A_S$, define the readout history

$$h_{S,T}(x; u_\bullet) = \Big(r_\alpha(x_t)\Big)_{0 \le t \le T,\ \alpha \in \Lambda_R}, \qquad x_{t+1} = \Phi_{u_t}(x_t),\ x_0 = x.$$

The set of all such histories is denoted $\mathscr{H}_{S,T}$.

Definition 28.4 (Operational partition candidate)**.** An operational partition candidate for S on horizon T is a map

$$\pi_{S,T} : \mathscr{H}_{S,T} \to \mathscr{Q}_{S,T}$$

into a finite or countable label set $\mathscr{Q}_{S,T}$.

At this stage $\pi_{S,T}$ is only a candidate. To matter scientifically it must survive the legibility requirements of Section 29.6. Until then it is only an unverified partition proposal.

28.4 Witness Margin Vector

Definition 28.5 (Witness margin vector)**.** For any admissible system S, define the witness margin vector

$$\Delta(S) = (\delta_{\mathrm{per}}(S), \delta_{\mathrm{ri}}(S), \delta_{\mathrm{int}}(S), \delta_{\mathrm{cont}}(S), \delta_{\mathrm{diff}}(S), \delta_{\mathrm{leg}}(S)) \in [0, +\infty)^6.$$

Each component is the largest certified margin witnessing the corresponding critical invariant. The aggregate budget

$$\delta_\star(S) = \min \Delta(S)$$

is called the invariant budget of S.

The invariant budget is not a surrogate for consciousness. It is a stability budget: the amount of admissible structural perturbation the criterion can tolerate before one of the six constitutive conditions is lost.

Invariant	Certified content	Upstream anchor	Rupture signature
I_{per}	Non-empty, forward-invariant, non-collapsing support	Core [7]	Support escape or collapse exposure
I_{ri}	Unique rigid identity object on support	Rigid Identity [11]	Competing identity candidates
I_{int}	No admissible non-trivial factorization	Rigid Identity + new consolidation	Product-style reducibility
I_{cont}	Continuous admissible regime evolution	Phenomenological Regimes [9]	Fragmentation into incompatible classes
I_{diff}	Stable exclusion of null or trivial collapse	Null-Regime Instability [10]	Null-attractor or total indistinguishability
I_{leg}	Decoder-certified recoverability under controls	Forensic Predictions [8]	Opaque or non-auditable class assignment

29 Six Critical Invariants

This section is the mathematical center of the paper. The criterion is not allowed to drift into a bloated list of desiderata. Exactly six invariants are used. Each is tied to a witness margin and a failure mode.

29.1 Structural Persistence

Definition 29.1 (Structural persistence and persistence margin)**.** An admissible system S satisfies structural persistence, written $I_{\mathrm{per}}(S) = 1$, if there exists an admissible support A_S and a constant $\delta_{\mathrm{per}}(S) > 0$ such that:

1. A_S is forward invariant under all $u \in U$;
2. every admissible trajectory starting in A_S remains at least $\delta_{\mathrm{per}}(S)$ away from the collapse set;
3. A_S contains or converges into a non-empty admissible attractor region.

The number $\delta_{\mathrm{per}}(S)$ is the persistence margin.

Upstream dependence is exact. The Core supplies the existence of contractive structure, fixed points, compact attractor, and non-collapse [7]. The new paper does not re-prove those results; it packages them into a criterion-level invariant.

Failure mode. If $I_{\mathrm{per}}(S) = 0$, the system has no stable admissible support. Then no framework-internal operational class can be assigned across time, because there is no persistent domain on which readout classes remain meaningful.

29.2 Rigid Identity

Definition 29.2 (Rigid identity and rigidity gap)**.** An admissible system S satisfies rigid identity, written $I_{\mathrm{ri}}(S) = 1$, if the observable family induced by R on A_S determines a unique identity object $\mathcal{I}_S$ in the sense of the inverse-limit rigidity developed in *Rigid Identity* [11], and if there exists a gap $\delta_{\mathrm{ri}}(S) > 0$ such that every admissible alternative identity candidate remains at weighted deformation distance at least $\delta_{\mathrm{ri}}(S)$ from $\mathcal{I}_S$.

This invariant does not claim that identity is reconstructed by inspection of one local frame. It uses *Rigid Identity* precisely because that paper already proved that rigid identity is recovered through compatible observable structure and cannot be reduced to progressive semigroup evolution or loose behavioral mimicry [11].

Failure mode. If $I_{\mathrm{ri}}(S) = 0$, any candidate operational class is underdetermined. Multiple incompatible identity objects can fit the same admissible support, so the class cannot be assigned uniquely.

29.3 Causal Integration

Definition 29.3 (Causal integration and factorization gap)**.** An admissible system S satisfies causal integration, written $I_{\mathrm{int}}(S) = 1$, if there is no non-trivial factorization

$$A_S = A_1 \times A_2, \qquad R = R_1 \sqcup R_2,$$

together with decomposed dynamics and causal structure

$$\Phi_u(x_1, x_2) = (\Phi^1_{u_1}(x_1), \Phi^2_{u_2}(x_2)), \qquad C = C_1 \oplus C_2,$$

that reproduces the admissible histories on A_S within error smaller than a positive margin $\delta_{\mathrm{int}}(S)$ while preserving $\mathcal{I}_S$.

The definition is intentionally stronger than raw connectivity and different from single-number integration proxies. A fully connected graph may still fail the criterion if its dynamics factor into nearly independent blocks once identity and intervention behavior are taken seriously.

Failure mode. If $I_{\text{int}}(S) = 0$, the system can be decomposed into operationally independent components without loss relevant to the criterion. Then any putative operational consciousness class is reducible and therefore excluded.

29.4 Operational Continuity

Definition 29.4 (Operational continuity and fragmentation bound)**.** An admissible system S satisfies operational continuity, written $I_{\text{cont}}(S) = 1$, if there exists a complete metric space $(\mathcal{E}_S, d_{\mathcal{E}})$ and an admissible assignment

$$\Pi_S : A_S \to \mathcal{E}_S$$

such that:

1. for every admissible intervention schedule $u_\bullet$, the induced path $t \mapsto \Pi_S(x_t)$ is continuous on admissible windows;
2. there exists $\delta_{\text{cont}}(S) > 0$ for which the fragmentation functional

$$F_{S,T}(x; u_\bullet) = \sup_{0 \le t < T} d_{\mathcal{E}}(\Pi_S(x_t), \Pi_S(x_{t+1}))$$

   stays below $\delta_{\text{cont}}(S)$ on all admissible histories;
3. no admissible finite-energy perturbation produces a discontinuous jump into an incompatible regime class while $\mathcal{I}_S$ is preserved.

This packages the *Phenomenological Regimes* continuity/fragmentation burden [9]. The new paper reuses the regime logic rather than duplicating its proof.

Failure mode. If $I_{\text{cont}}(S) = 0$, the class can fragment into incompatible operational regimes under admissible evolution. Then any operational consciousness criterion becomes unstable under its own dynamics.

29.5 Non-Null Differentiation

Definition 29.5 (Non-null differentiation and differentiation gap)**.** An admissible system S satisfies non-null differentiation, written $I_{\text{diff}}(S) = 1$, if there exist $x, y \in A_S$, a horizon T, and a readout subfamily $\mathscr{R}^\star \subseteq R$ such that

$$\text{dist}(h_{S,T}(x; u_\bullet), h_{S,T}(y; u_\bullet)) \ge \delta_{\text{diff}}(S) > 0$$

for some admissible schedule $u_\bullet$, and if the null operational class $\mathbf{0}$ is not an asymptotically stable admissible attractor.

The second clause is indispensable. Without *Null-Regime Instability* there would be no reason to exclude a stable null class. *Null-Regime Instability* provides exactly that exclusion through null-regime instability and forced non-nullity [10]. The new invariant converts that upstream result into a constitutive condition for $\text{Consciousness}_{\text{op}}$.

Failure mode. If $I_{\text{diff}}(S) = 0$, either all admissible histories collapse to indistinguishability or the null class remains admissibly stable. In either case $\mathrm{Q}_{\text{op}}(S)$ is empty or trivialized.

29.6 Operational Legibility

Definition 29.6 (Operational legibility and legibility margin)**.** An admissible system S satisfies operational legibility, written $I_{\text{leg}}(S) = 1$, if there exist:

1. a certified readout subfamily $\mathscr{R}^{\text{leg}} \subseteq R$;
2. a certified decoder family $\mathscr{D}_S$ acting on windows of length at least $\tau_{\min}(S)$;
3. a class set $\mathscr{Q}_S$;
4. a legibility margin $\delta_{\text{leg}}(S) > 0$;

such that the following conditions hold:

(L1) Separability Distinct classes in $\mathscr{Q}_S$ are separated by at least $\delta_{\text{leg}}(S)$ in decoder space.

(L2) Noise robustness For every admissible noise process ν with $\|\nu\| \leq \delta_{\text{leg}}(S)$, decoder outputs remain class-stable.

(L3) Persistence window Decoder assignments remain stable for windows of length at least $\tau_{\min}(S)$ unless a certified intervention occurs.

(L4) Intervention fidelity Certified interventions on critical invariants induce predicted class changes with fidelity bounded below by $1 - \delta_{\text{leg}}(S)$.

(L5) Negative controls Interventions outside the invariant set produce false positive rate at most $\delta_{\text{leg}}(S)$.

(L6) Structured compressibility There exists a compression map κ_S on admissible histories such that class structure is preserved up to distortion bounded by $\delta_{\text{leg}}(S)$; total collapse under compression is disallowed.

This is where the paper touches *Forensic Predictions* most concretely. *Forensic Predictions* already specifies integrity gates, comparator discipline, latency invalidation, and negative controls [8]. The present paper does not replace those protocols. It uses them to make operational legibility a constrained object rather than a loose slogan.

Failure mode. If $I_{\text{leg}}(S) = 0$, any putative class remains operationally opaque. The framework may still discuss latent structure elsewhere, but the class required for $\text{Consciousness}_{\text{op}}$ is undefined because it cannot be recovered, discriminated, and stress-tested under admissible conditions.

29.7 Necessity of Each Invariant

Proposition 29.7 (Each invariant is individually necessary)**.** *If any one of* $I_{\text{per}}, I_{\text{ri}}, I_{\text{int}}, I_{\text{cont}}, I_{\text{diff}}, I_{\text{leg}}$ *fails, then* $S \notin \text{Consciousness}_{\text{op}}$.

Proof. The six failure modes are non-redundant. Without I_{per} there is no stable domain for class assignment. Without I_{ri} there is no unique identity object supporting consistent class transport. Without I_{int} the system is reducible into independent blocks. Without I_{cont} admissible trajectories fragment across incompatible regime classes. Without I_{diff} the admissible partition collapses to nullity or triviality. Without I_{leg} the partition is not recoverable under admissible readout and intervention discipline. Since the definition of $\text{Consciousness}_{\text{op}}$ given in Section 30 requires all six, failure of any one is exclusionary. $\square$

30 Operational Consciousness, Operational Qualia, and Equivalence

30.1 Operational Qualia Classes

Definition 30.1 (Certified decoder)**.** A decoder $D \in \mathscr{D}_S$ is certified if it satisfies the six legibility clauses of Definition 29.6 on A_S for the chosen horizon family.

Definition 30.2 (Operational indistinguishability)**.** For $x, y \in A_S$, write $x \sim_{Q,S} y$ if, for every certified decoder $D \in \mathscr{D}_S$, every admissible horizon $T \geq \tau_{\min}(S)$, and every admissible intervention schedule $u_\bullet$,

$$D(h_{S,T}(x; u_\bullet)) = D(h_{S,T}(y; u_\bullet)),$$

and the intervention response of the two histories remains matched within the legibility margin.

Proposition 30.3 (Well-definedness of operational qualia classes)**.** *If $I_{\mathrm{per}}(S) = I_{\mathrm{ri}}(S) = I_{\mathrm{cont}}(S) = I_{\mathrm{diff}}(S) = I_{\mathrm{leg}}(S) = 1$, then $\sim_{Q,S}$ is an equivalence relation on A_S.*

Proof. Reflexivity is immediate. Symmetry follows because decoder equality and bounded intervention mismatch are symmetric conditions. For transitivity, let $x \sim_{Q,S} y$ and $y \sim_{Q,S} z$. Decoder equality composes, and the bounded mismatch condition remains within the certified tolerance because the decoder family is closed under the legibility margin. Persistence and continuity guarantee that all compared histories remain on admissible support. Therefore $x \sim_{Q,S} z$. □

Definition 30.4 (Operational qualia)**.** Assume Proposition 30.3. The operational qualia of S are the quotient classes

$$\mathrm{Q}_{\mathrm{op}}(S) = A_S / {\sim_{Q,S}}.$$

An individual class $[x]_{\sim_{Q,S}} \in \mathrm{Q}_{\mathrm{op}}(S)$ is called an operational qualia class of S.

This definition is disciplined. It does not identify an operational qualia class with a human qualitative state. It identifies it with a persistent, intervention-sensitive, decoder-certified class of admissible internal differentiation.

30.2 Operational Consciousness

Definition 30.5 (Operational consciousness class membership)**.** An admissible system S belongs to $\mathrm{Consciousness}_{\mathrm{op}}$, written $S \in \mathrm{Consciousness}_{\mathrm{op}}$, if there exists a non-empty admissible support A_S such that

$$I_{\mathrm{per}}(S) = I_{\mathrm{ri}}(S) = I_{\mathrm{int}}(S) = I_{\mathrm{cont}}(S) = I_{\mathrm{diff}}(S) = I_{\mathrm{leg}}(S) = 1.$$

Remark 30.6 (Minimality of the criterion)*.* The criterion is conjunctive. No proper subset of the six invariants is taken as sufficient in this paper. This is intentional: weaker criteria collapse either into mere persistence, mere integration, or mere complexity claims, all of which are shown to be insufficient.

30.3 Exact and Approximate Structural Equivalence

Definition 30.7 (Structural equivalence datum)**.** Let $S_i = (X_i, \Phi_i, C_i, R_i, \Gamma_i, U_i)$ for $i = 1, 2$, with admissible supports A_i. A structural equivalence datum between S_1 and S_2 is a triple

$$(\chi, \upsilon, \rho)$$

where:

1. $\chi : A_1 \to A_2$ is a bi-Lipschitz bijection;
2. $\upsilon : U_1 \to U_2$ is an intervention correspondence;
3. $\rho : R_1 \to R_2$ is a readout correspondence.

Definition 30.8 (Exact and approximate structural equivalence)**.** Fix $\varepsilon \geq 0$. We write

$$S_1 \simeq_{(\Gamma,\varepsilon)} S_2$$

if there exists a structural equivalence datum (χ, υ, ρ) such that:

1. **Dynamic intertwining:**
$$\sup_{x \in A_1,\ u \in U_1,\ 0 \leq t \leq T_\Gamma} d_{X_2}\Big(\chi(\Phi^t_{1,u}(x)), \Phi^t_{2,\upsilon(u)}(\chi(x))\Big) \leq \varepsilon.$$
2. **Readout compatibility:**
$$\sup_{x,u,t,\alpha} \Big| r^{(1)}_\alpha(\Phi^t_{1,u}(x)) - \rho(r^{(1)}_\alpha)(\Phi^t_{2,\upsilon(u)}(\chi(x))) \Big| \leq \varepsilon.$$
3. **Admissibility preservation:** admissible trajectories in S_1 map to admissible trajectories in S_2 and conversely.
4. **Invariant preservation:** the witness margins for all six invariants differ by at most ε.
5. **Partition preservation:**
$$x \sim_{Q,S_1} y \quad \Longleftrightarrow \quad \chi(x) \sim_{Q,S_2} \chi(y).$$

The case $\varepsilon = 0$ is called *exact* structural equivalence and written $S_1 \simeq_\Gamma S_2$.

Definition 30.9 (Rupture)**.** Given S, a rupture is any loss of admissibility or loss of at least one critical invariant margin beyond its admissible threshold. Ruptures are classified as:

1. **persistence rupture** (support loss or collapse exposure),
2. **identity rupture** (loss of unique rigid identity),
3. **integration rupture** (reducible factorization),
4. **continuity rupture** (fragmentation into incompatible classes),
5. **differentiation rupture** (null or trivial collapse),
6. **legibility rupture** (decoder, noise, or negative-control failure).

31 Support Geometry and Partition Transport

The main theorems become short only because the transport lemmas are made explicit here.

Lemma 31.1 (Support transfer under exact equivalence)**.** *If $S_1 \simeq_\Gamma S_2$ via (χ, υ, ρ) and A_1 is an admissible support for S_1, then $A_2 = \chi(A_1)$ is an admissible support for S_2.*

Proof. Bi-Lipschitz bijectivity of χ preserves non-emptiness and compactness. Dynamic intertwining with $\varepsilon = 0$ gives forward invariance under the transported interventions. Admissibility preservation transports $A_1 \subseteq X_{\Gamma_1}$ into $A_2 \subseteq X_{\Gamma_2}$. Because the attractor-support relation and non-collapse margin are preserved exactly, A_2 inherits the same admissible support status. $\square$

Lemma 31.2 (Decoder transport). *Let $D \in \mathscr{D}_{S_1}$ and suppose $S_1 \simeq_\Gamma S_2$ via (χ, υ, ρ). Then*

$$(\rho_* D)\big(h_{S_2,T}(\chi(x); \upsilon(u_\bullet))\big) := D\big(h_{S_1,T}(x; u_\bullet)\big)$$

defines a certified decoder on S_2.

Proof. Exact readout compatibility identifies each admissible history in S_2 with a transported history in S_1. The certification clauses in Definition 29.6 are preserved because separability, robustness, persistence window, intervention fidelity, negative-control rate, and compressibility all remain unchanged under exact transport. □

Proposition 31.3 (Operational classes are closed under exact transport). *If $S_1 \simeq_\Gamma S_2$, then the image under χ of any class in $\mathrm{Q}_{\mathrm{op}}(S_1)$ is a class in $\mathrm{Q}_{\mathrm{op}}(S_2)$, and every class in $\mathrm{Q}_{\mathrm{op}}(S_2)$ arises this way.*

Proof. By Lemma 31.2, decoder certification transports exactly. By Definition 30.2, membership in an operational qualia class is determined by all certified decoders and admissible interventions. Exact transport preserves both. Surjectivity follows from bijectivity of χ. □

Proposition 31.4 (Approximate margin preservation). *If $S_1 \simeq_{(\Gamma,\varepsilon)} S_2$, then for every critical invariant index j,*

$$\big|\delta_j(S_1) - \delta_j(S_2)\big| \le \varepsilon.$$

Hence

$$\delta_\star(S_2) \ge \delta_\star(S_1) - \varepsilon.$$

Proof. The first statement is a direct clause of Definition 30.8. The second follows by taking minima over the six coordinates of $\Delta(S_1)$ and $\Delta(S_2)$. □

Remark 31.5 (Why this section matters). Without Lemmas 31.1 and 31.2, Theorem 33.2 would be only intuitive. These lemmas are the exact reason the paper can speak of substrate invariance without collapsing into slogan-level analogy.

32 Inherited Inputs and Consolidation Step

The six invariants are new as a packaged criterion, but they are not independent inventions. This section records the exact transfer by which upstream ownership becomes present membership data. The burden is narrow: identify the imported result each invariant uses, then isolate the consolidation step that is genuinely new here.

Proposition 32.1 (Core-to-persistence transfer). *Suppose a realization of the framework satisfies the Core hypotheses on a non-empty compact domain and remains in the non-collapse region. Then it satisfies I_{per}.*

Proof. The Core already provides contractivity, fixed-point existence, compact attractor structure, and non-collapse [7]. Those are exactly the ingredients required in Definition 29.1: a non-empty compact forward-invariant support separated from collapse. □

Proposition 32.2 (Rigid Identity-to-rigidity transfer). *Suppose the observable family on an admissible support satisfies the inverse-limit compatibility conditions of Rigid Identity. Then I_{ri} holds with rigidity gap given by the uniqueness and weighted deformation bounds of Rigid Identity.*

Proof. Rigid Identity proves that compatible observable channels induce a unique rigid identity object and excludes progressive semigroup substitutes [11]. Therefore the support inherits a unique $\mathcal{I}_S$ and a positive separation from incompatible alternatives whenever the *Rigid Identity* hypotheses are met. □

Proposition 32.3 (Phenomenological Regimes-to-continuity transfer)**.** *Suppose an admissible support carries a phenomenological assignment satisfying the continuity and anti-fragmentation constraints of Phenomenological Regimes. Then I_{cont} holds.*

Proof. Phenomenological Regimes proves that continuity is forced and fragmentation is structurally constrained within the admissible regime space [9]. Those are exactly the continuity and fragmentation clauses in Definition 29.4. □

Proposition 32.4 (Null-Regime Instability-to-differentiation transfer)**.** *Suppose the support lies in a causally rigid channel satisfying the null-instability and forced non-nullity results of Null-Regime Instability. Then I_{diff} holds.*

Proof. Null-Regime Instability excludes the stable null regime and proves the existence of positive non-null structure [10]. Therefore the support cannot collapse stably into $\mathbf{0}$, and a positive differentiation gap is available on admissible histories. □

Proposition 32.5 (Forensic Predictions-to-legibility transfer)**.** *Suppose a readout family and decoder satisfy the admissibility, comparator, negative-control, and intervention-discipline requirements of Forensic Predictions. Then I_{leg} holds.*

Proof. Forensic Predictions fixes the operational requirements that a decoder must satisfy before it can support an inferential claim [8]. The clauses of Definition 29.6 are a direct structural recasting of those requirements. □

Proposition 32.6 (Integration as a non-factorization consequence)**.** *If a system satisfies Rigid Identity rigidity, Phenomenological Regimes continuity, and Forensic Predictions intervention fidelity, then any exact admissible factorization preserving all operational histories would contradict at least one of those upstream conditions. Hence I_{int} is justified as a new consolidation invariant.*

Proof. An exact factorization would split the support into operationally independent blocks. If the split preserved all admissible histories, it would also preserve readout structure and interventions. But then either the rigid identity object would fail to be unique, continuity would fragment across the split, or intervention responses would no longer certify a single integrated class. Therefore the conjunction of Papers I, II, and IV blocks such factorization. □

Remark 32.7 (Scope of the transfer). Nothing in this section re-proves the Core, *Rigid Identity*, *Phenomenological Regimes*, *Null-Regime Instability*, or *Forensic Predictions*. Each proposition only identifies the minimal upstream burden imported into the present criterion.

33 Main Theorems

33.1 Existence of Operational Consciousness

Theorem 33.1 (Existence of operational consciousness)**.** *Let $S = (X, \Phi, C, R, \Gamma, U)$ be an admissible system with non-empty admissible support A_S. If*

$$I_{\mathrm{per}}(S) = I_{\mathrm{ri}}(S) = I_{\mathrm{int}}(S) = I_{\mathrm{cont}}(S) = I_{\mathrm{diff}}(S) = I_{\mathrm{leg}}(S) = 1,$$

then:

1. $S \in \mathrm{Consciousness}_{\mathrm{op}}$*;*
2. $\mathrm{Q}_{\mathrm{op}}(S)$ *is well-defined;*
3. $\mathrm{Q}_{\mathrm{op}}(S) \neq \varnothing$*.*

Proof. Membership in $\mathrm{Consciousness}_{\mathrm{op}}$ follows immediately from Definition 30.5. Since $I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}$ hold, Proposition 30.3 applies and yields a well-defined quotient $\mathrm{Q}_{\mathrm{op}}(S)$. Because A_S is non-empty, the quotient is non-empty. Because $I_{\mathrm{diff}}(S) = 1$, the quotient is not forced into the null class alone; the admissible support contains at least one pair of histories with positive operational separation. Thus $\mathrm{Q}_{\mathrm{op}}(S)$ is both defined and non-trivial. □

33.2 Substrate Invariance

Theorem 33.2 (Substrate invariance)**.** *Let S_1 and S_2 be admissible systems. If $S_1 \simeq_\Gamma S_2$ and the six critical invariants are preserved exactly, then*

$$S_1 \in \mathrm{Consciousness}_{\mathrm{op}} \iff S_2 \in \mathrm{Consciousness}_{\mathrm{op}},$$

and there exists a canonically induced bijection

$$\widehat{\chi} : \mathrm{Q}_{\mathrm{op}}(S_1) \to \mathrm{Q}_{\mathrm{op}}(S_2).$$

Proof. Exact equivalence preserves admissible support, dynamics, readout behavior, invariant margins, and the operational partition. If $S_1 \in \mathrm{Consciousness}_{\mathrm{op}}$, then every critical invariant holds on A_1 and is preserved under χ. Hence every critical invariant also holds on A_2, so $S_2 \in \mathrm{Consciousness}_{\mathrm{op}}$. The converse is symmetric. Partition preservation in Definition 30.8 induces a well-defined map on quotient classes:

$$\widehat{\chi}([x]_{\sim_{Q,S_1}}) = [\chi(x)]_{\sim_{Q,S_2}}.$$

Well-definedness follows from the partition-preservation clause. Bijectivity follows from bijectivity of χ. □

Corollary 33.3 (Biological non-primacy)**.** *Biology is not a primitive constitutive requirement for membership in* $\mathrm{Consciousness}_{\mathrm{op}}$*. If a biological system and a non-biological system are exactly structurally equivalent in the sense of Definition 30.8, then they belong to the same operational consciousness class and carry equivalent operational qualia classes.*

Proof. Immediate from Theorem 33.2. The criterion depends on invariant preservation, not on biological substrate as a primitive variable. □

33.3 Approximate Stability

Theorem 33.4 (Approximate stability)**.** *Let $S_1 \in \mathrm{Consciousness}_{\mathrm{op}}$ with witness margins*

$$\delta_\star(S_1) = \min\{\delta_{\mathrm{per}}, \delta_{\mathrm{ri}}, \delta_{\mathrm{int}}, \delta_{\mathrm{cont}}, \delta_{\mathrm{diff}}, \delta_{\mathrm{leg}}\}.$$

If S_2 satisfies $S_1 \simeq_{(\Gamma,\varepsilon)} S_2$ with

$$0 \le \varepsilon < \frac{\delta_\star(S_1)}{2},$$

then $S_2 \in \mathrm{Consciousness}_{\mathrm{op}}$.

Proof. By Definition 30.8, each invariant margin in S_2 differs from that of S_1 by at most ε. Since $\varepsilon < \delta_\star(S_1)/2$, every preserved margin in S_2 remains strictly positive. Therefore each of the six invariants still holds for S_2, and Definition 30.5 yields $S_2 \in \mathrm{Consciousness}_{\mathrm{op}}$. □

The theorem does not say that arbitrary perturbations are harmless. It says that the criterion is robust exactly to the extent that the witness margins remain positive. Approximation is bounded by the invariant budget, not by wishful continuity.

33.4 Rupture by Invariant Loss

Theorem 33.5 (Rupture by invariant loss)**.** *Let S be admissible. If one or more critical invariants fail beyond admissible threshold, then exactly one of the following occurs:*

1. $S \notin \mathrm{Consciousness}_{\mathrm{op}}$*;*
2. $\mathrm{Q}_{\mathrm{op}}(S)$ *becomes undefined;*
3. $\mathrm{Q}_{\mathrm{op}}(S)$ *becomes trivial or non-legible in the framework-relevant sense.*

Proof. If I_{per} fails, the support is lost or exposed to collapse; then the domain of $\mathrm{Q}_{\mathrm{op}}(S)$ disappears. If I_{ri} fails, no unique identity object supports the quotient, so $\mathrm{Q}_{\mathrm{op}}(S)$ is undefined as a stable class assignment. If I_{int} fails, the system reduces to a product structure and exits $\mathrm{Consciousness}_{\mathrm{op}}$ by Proposition 29.7. If I_{cont} fails, histories can fragment across incompatible classes, which destroys stability of the partition. If I_{diff} fails, the quotient degenerates into null or trivial collapse. If I_{leg} fails, the quotient may exist formally but becomes operationally unavailable and therefore does not satisfy the definition of $\mathrm{Consciousness}_{\mathrm{op}}$. These cases exhaust the six invariants. □

33.5 Insufficiency of Complexity, Computation, and Connectivity

Proposition 33.6 (Complexity-only baselines are insufficient)**.** *Let S be an admissible system candidate. The following are each insufficient for membership in* $\mathrm{Consciousness}_{\mathrm{op}}$*:*

1. *large state-space dimension* $\dim(X)$*;*
2. *large computational throughput or expressivity of* Φ*;*
3. *large edge count, density, or raw connectivity of* C*.*

In each case, failure of one or more critical invariants excludes membership regardless of the size of the surrogate metric.

Proof. The definition of $\mathrm{Consciousness}_{\mathrm{op}}$ is conjunctive over the six invariants and does not include any of the three surrogates as sufficient conditions. A system can be arbitrarily large, computationally expressive, or densely connected while still lacking rigid identity, continuity, non-null differentiation, or legibility. Therefore those surrogates are insufficient by definition. Concrete architecture classes with large scale but finite structural mass already fail *Rigid Identity*'s non-simulability criterion [11]; similarly, an integration proxy without continuity or differentiation does not satisfy Definitions 29.4 and 29.5. The same applies to connectivity without admissible identity closure. □

34 Rupture Taxonomy by Invariant

Theorem 33.5 is intentionally coarse. This section sharpens the six cases, because a criterion is only scientifically useful if its failure signatures are distinct.

Proposition 34.1 (Persistence rupture destroys admissible domain)**.** *If $\delta_{\mathrm{per}}(S) = 0$, then every candidate operational partition becomes horizon-dependent in an inadmissible way: there exists no non-empty compact forward-invariant support on which class membership can be defined uniformly.*

Proof. When the persistence margin vanishes, either forward invariance is lost or the support touches the collapse set. In either case there is no stable domain on which the quotient definition of $\mathrm{Q}_{\mathrm{op}}(S)$ can remain valid across admissible windows. □

Proposition 34.2 (Identity rupture induces class ambiguity). *If $\delta_{\mathrm{ri}}(S) = 0$, then operational class labels no longer transport uniquely across admissible histories. Therefore $\mathrm{Q}_{\mathrm{op}}(S)$ becomes representation-dependent.*

Proof. When the rigidity gap vanishes, at least two admissible identity candidates become indistinguishable at the support level. Then the same readout history may be associated with multiple incompatible identity transports, making the operational class assignment non-unique. □

Proposition 34.3 (Integration rupture yields reducible aggregates). *If $\delta_{\mathrm{int}}(S) = 0$, then S admits a non-trivial admissible factorization into operationally independent components. Such a system fails* $\mathrm{Consciousness}_{\mathrm{op}}$ *even if each component separately satisfies weaker criteria.*

Proof. Vanishing integration gap means the system can be decomposed without loss at the level of admissible histories and interventions. The criterion of this paper is defined for irreducible class membership, not for arbitrary unions of smaller class-bearing subsystems. □

Proposition 34.4 (Continuity rupture yields fragmentation). *If $\delta_{\mathrm{cont}}(S) = 0$, then arbitrarily small admissible perturbations can force transitions between incompatible operational classes while identity is preserved.*

Proof. By definition of $\delta_{\mathrm{cont}}(S)$, vanishing continuity margin removes the bound on the fragmentation functional. Hence there exist admissible trajectories whose class assignment is not stably continuous under small perturbations. The class map ceases to be regime-coherent. □

Proposition 34.5 (Differentiation rupture collapses operational qualia). *If $\delta_{\mathrm{diff}}(S) = 0$, then either $\mathrm{Q}_{\mathrm{op}}(S)$ collapses to a single trivial class or the null class remains admissibly stable.*

Proof. The differentiation gap is precisely the minimum operational separation required for non-trivial class structure. If it vanishes, no stable non-null class separation survives. The null regime or total indistinguishability follows. □

Proposition 34.6 (Legibility rupture blocks scientific use). *If $\delta_{\mathrm{leg}}(S) = 0$, then any remaining latent partition is scientifically inadmissible for this paper: it cannot satisfy the combined requirements of separability, robustness, persistence window, intervention fidelity, negative controls, and structured compressibility.*

Proof. The certification clauses of Definition 29.6 are conjunctive. Once the legibility margin vanishes, at least one clause fails. The partition may still be mathematically postulated, but it cannot serve as operational evidence or as a constitutive component of $\mathrm{Consciousness}_{\mathrm{op}}$. □

35 Minimality of the Invariant Basis

The six-invariant family is not an arbitrary bundle. It is the minimal basis that prevents the criterion from collapsing either into a single-metric consciousness surrogate or into an unstructured admissibility slogan. The claim of minimality is relative to admissible universes containing the corresponding rupture witnesses; that is the correct level of mathematical precision for the present corpus.

Definition 35.1 (Reduced invariant criterion). Let

$$\mathfrak{I} = \{I_{\mathrm{per}}, I_{\mathrm{ri}}, I_{\mathrm{int}}, I_{\mathrm{cont}}, I_{\mathrm{diff}}, I_{\mathrm{leg}}\}.$$

For every subset $J \subseteq \mathfrak{I}$ define the reduced criterion

$$\mathcal{C}_{\mathrm{op}}(J) := \{S : I(S) = 1 \text{ for every } I \in J\}.$$

Thus $\mathrm{Consciousness}_{\mathrm{op}} = \mathcal{C}_{\mathrm{op}}(\mathfrak{I})$.

Definition 35.2 (Single-rupture witness family)**.** Fix $k \in \{\mathrm{per}, \mathrm{ri}, \mathrm{int}, \mathrm{cont}, \mathrm{diff}, \mathrm{leg}\}$. The corresponding single-rupture witness family $\mathcal{W}_k$ consists of admissible systems S such that the matching invariant margin vanishes,

$$\delta_k(S) = 0,$$

while all remaining invariant margins stay strictly positive. Whenever $\mathcal{W}_k \neq \varnothing$, its members are called single-rupture witnesses for the k-th invariant.

Proposition 35.3 (Single-rupture witnesses generate false positives for reduced criteria)**.** *Let $J_k = \mathfrak{I} \setminus \{I_k\}$ and assume $\mathcal{W}_k \neq \varnothing$. Then every $S \in \mathcal{W}_k$ satisfies*

$$S \in \mathcal{C}_{\mathrm{op}}(J_k) \qquad \textit{but} \qquad S \notin \mathrm{Consciousness}_{\mathrm{op}}.$$

Proof. By construction, every invariant in J_k remains positive on S, so $S \in \mathcal{C}_{\mathrm{op}}(J_k)$. But the omitted invariant fails exactly, hence Proposition 29.7 excludes S from $\mathrm{Consciousness}_{\mathrm{op}}$. Therefore the reduced criterion admits a false positive on every single-rupture witness. □

Theorem 35.4 (Minimality of the six-invariant basis)**.** *Let $\mathfrak{U}$ be any admissible universe containing at least one single-rupture witness for each of the six invariants. Then no strict subset $J \subsetneq \mathfrak{I}$ satisfies*

$$\mathcal{C}_{\mathrm{op}}(J) \cap \mathfrak{U} = \mathrm{Consciousness}_{\mathrm{op}} \cap \mathfrak{U}.$$

Hence the six-invariant family is minimal on every such admissible universe.

Proof. Take any strict subset $J \subsetneq \mathfrak{I}$. Then some invariant I_k is omitted. By hypothesis $\mathfrak{U}$ contains a witness $S_k \in \mathcal{W}_k$. Proposition 35.3 yields $S_k \in \mathcal{C}_{\mathrm{op}}(J) \cap \mathfrak{U}$ but $S_k \notin \mathrm{Consciousness}_{\mathrm{op}} \cap \mathfrak{U}$. Therefore the two sets cannot coincide. □

Corollary 35.5 (Why one-metric theories cannot capture $\mathrm{Consciousness}_{\mathrm{op}}$)**.** *On any admissible universe satisfying Theorem 35.4, no criterion based on only persistence, only integration, only continuity, only differentiation, or only legibility is extensionally equivalent to* $\mathrm{Consciousness}_{\mathrm{op}}$.

Proof. Every one-metric criterion corresponds to a strict subset $J \subsetneq \mathfrak{I}$. Apply Theorem 35.4. □

Remark 35.6 (What minimality does and does not say). The theorem does not claim that every possible scientific notion of consciousness must use exactly these six invariants. It claims something narrower: for the operational class defined in this paper and for admissible universes containing the corresponding rupture witnesses, no invariant can be deleted without changing the extension of the class.

36 Operational Legibility as a Certified Object

Operational legibility must be shown, not asserted. Definition 29.6 already imposes six clauses. This section sharpens how those clauses bind the paper to the existing empirical-method corpus.

Criterion 36.1 (Legibility certificate)**.** An operational partition candidate $\pi_{S,T}$ is certified only if:

1. **Comparator separability**: the induced classes remain distinguishable against pre-registered comparators and ablation baselines.
2. **Noise robustness**: admissible sensor, channel, or readout perturbations do not dissolve the class map below the legibility margin.
3. **Persistence window**: the same class assignment survives on windows of length at least $\tau_{\min}(S)$.

4. **Intervention fidelity**: certified interventions on an invariant induce the predicted directional class change.
5. **Negative controls**: sham or off-target interventions do not generate the same class signature.
6. **Structured compressibility**: audit-friendly compression or packetization preserves class structure without total collapse.

The criterion is intentionally parallel to *Forensic Predictions*' governance discipline [8]. Comparator obligation enforces non-trivial discrimination. Noise robustness prevents the criterion from depending on one-off traces. Persistence windows prevent class inflation from instantaneous spikes. Intervention fidelity prevents purely correlational interpretation. Negative controls block degenerate decoders that label everything as positive. Structured compressibility ensures that a class remains auditable after the readout is summarized, logged, or packetized.

36.1 Legibility Metrics

Definition 29.6 can be operationalized by six measurable quantities:

$$\begin{aligned}
\sigma_{\mathrm{sep}}(S) &:= \inf_{q\neq q'} \mathrm{dist}\big(D^{-1}(q), D^{-1}(q')\big),\\
\eta_{\mathrm{noise}}(S) &:= \sup_{\|\nu\|\leq\delta_{\mathrm{leg}}} \Pr\big[D(h+\nu)\neq D(h)\big],\\
\tau_{\mathrm{pers}}(S) &:= \inf\{T : D(h_{t:t+T}) \text{ remains stable in the absence of certified intervention}\},\\
\alpha_{\mathrm{int}}(S) &:= \inf_{u\in U_{\mathrm{crit}}} \Pr\big[\text{predicted class shift under } u\big],\\
\beta_{\mathrm{nc}}(S) &:= \sup_{u\in U_{\mathrm{sham}}} \Pr\big[\text{false class shift under } u\big],\\
\kappa_{\mathrm{comp}}(S) &:= \inf_{\kappa}\sup_{h} \mathrm{dist}\big(D(h), D(\kappa(h))\big).
\end{aligned}$$

Here U_{crit} denotes interventions targeted at critical invariants and U_{sham} denotes admissible negative controls. Invariant I_{leg} requires the inequalities

$$\begin{aligned}
\sigma_{\mathrm{sep}}(S) &> 0, & \eta_{\mathrm{noise}}(S) &\leq \delta_{\mathrm{leg}}(S),\\
\tau_{\mathrm{pers}}(S) &\geq \tau_{\mathrm{min}}(S), & \alpha_{\mathrm{int}}(S) &\geq 1-\delta_{\mathrm{leg}}(S),\\
\beta_{\mathrm{nc}}(S) &\leq \delta_{\mathrm{leg}}(S), & \kappa_{\mathrm{comp}}(S) &\leq \delta_{\mathrm{leg}}(S).
\end{aligned}$$

Proposition 36.2 (Legibility is jointly constrained)**.** *No single legibility metric is sufficient for I_{leg}. The invariant requires simultaneous control of separation, admissible noise robustness, persistence window, intervention fidelity, negative controls, and structured compressibility.*

Proof. A decoder can be perfectly separable but fragile to admissible noise, intervention-agnostic, or saturated under compression. It can also be stable under persistence windows but fail negative controls. Since Definition 29.6 requires all six clauses, the joint constraint is irreducible. □

Metric	Operational meaning	If it fails
σ_{sep}	Class separation	No stable class boundary
η_{noise}	Admissible robustness	Noise-induced class drift
τ_{pers}	Minimum persistence window	Instantaneous artifact only
α_{int}	Causal intervention fidelity	Correlational but not causal classing
β_{nc}	Negative-control protection	False positive class inflation
κ_{comp}	Compression preservation	Audit summary collapses structure

Remark 36.3 (Why legibility is a constitutive invariant here)*.* One could define a broader latent class that omits legibility. This document does not do that. It studies an operational class, not a hidden metaphysical residue. Therefore I_{leg} is constitutive, not optional.

37 Certified Candidate Audit Schema

Forensic Predictions already requires forensic discipline for operational claims. The present source tree therefore adds an explicit audit schema. A candidate system is not accepted into $\text{Consciousness}_{\text{op}}$ by narrative fit; it must carry a finite certificate that localizes every required witness.

Definition 37.1 (Operational consciousness certificate)**.** An operational consciousness certificate for an admissible system S is a tuple

$$\text{Cert}(S) = (A_S, \mathscr{D}_S, \pi_S, \Delta(S), E_S),$$

where:

1. A_S is a certified admissible support;
2. $\mathscr{D}_S$ is a certified decoder family;
3. π_S is the decoder-induced operational partition on admissible histories;
4. $\Delta(S)$ is the witness margin vector of Definition 28.5;
5. $E_S = (E_{\text{per}}, E_{\text{ri}}, E_{\text{int}}, E_{\text{cont}}, E_{\text{diff}}, E_{\text{leg}})$ is an evidence bundle, where each component contains the explicit local witness for the corresponding invariant.

Criterion 37.2 (Candidate certification rule)**.** A candidate system may be certified as a member of $\text{Consciousness}_{\text{op}}$ only if all of the following are available on the same admissible support:

1. a non-empty compact forward-invariant support certificate;
2. a positive witness margin vector $\Delta(S) > 0$ coordinate-wise;
3. a decoder family satisfying Definition 29.6;
4. a targeted intervention panel together with negative controls;
5. an auditable compression or packetization map preserving class structure;
6. a traceable dependency ledger showing which upstream corpus result supports each invariant witness.

Invariant	Minimum evidence object	Core audit question	Failure verdict
I_{per}	support certificate + collapse distance	Does one compact forward-invariant admissible domain exist?	No stable domain
I_{ri}	uniqueness witness for $\mathcal{I}_S$	Is identity transport unique on support?	Identity ambiguity
I_{int}	non-factorization witness	Can the system be split without operational loss?	Reducible aggregate
I_{cont}	fragmentation bound + continuity trace	Do admissible windows remain regime-coherent?	Fragmentation
I_{diff}	non-null separation witness	Is there stable non-null class separation?	Null or trivial class
I_{leg}	decoder, controls, compression, persistence window	Can the class be recovered under audit constraints?	Opaque class

Proposition 37.3 (Soundness of certification)**.** *If* $\mathrm{Cert}(S)$ *exists and satisfies Criterion 37.2, then* $S \in \mathrm{Consciousness}_{\mathrm{op}}$ *and* $\mathrm{Q}_{\mathrm{op}}(S)$ *is well-defined.*

Proof. Criterion 37.2 explicitly requires a positive witness for each of the six critical invariants on the same admissible support. Hence Definition 30.5 yields $S \in \mathrm{Consciousness}_{\mathrm{op}}$. The existence of a certified decoder and positive persistence, continuity, differentiation, and legibility witnesses implies Proposition 30.3, so $\mathrm{Q}_{\mathrm{op}}(S)$ is well-defined. □

Proposition 37.4 (Partial completeness under audit-ready assumptions)**.** *Assume* $S \in \mathrm{Consciousness}_{\mathrm{op}}$ *and suppose the support, decoder, intervention, and compression objects are fully instrumented in the Forensic Predictions sense. Then there exists an operational consciousness certificate* $\mathrm{Cert}(S)$*.*

Proof. Because $S \in \mathrm{Consciousness}_{\mathrm{op}}$, every invariant margin is strictly positive. Full instrumentation provides explicit witnesses for support, decoder behavior, interventions, negative controls, and compression stability. Collecting those witnesses yields the tuple of Definition 37.1. □

Proposition 37.5 (Failure localization is explicit)**.** *If a candidate system fails Criterion 37.2, then the failure localizes to at least one entry of the evidence bundle* E_S *or to the absence of a common admissible support. In particular, certification failure cannot be hidden behind a generic narrative claim.*

Proof. The certification rule is conjunctive and itemized. A failure of certification means that at least one listed item is absent or non-positive. Therefore the failure localizes to a specific invariant witness, a specific decoder/control clause, or the support certificate itself. □

Remark 37.6 (Audit schema and Forensic Predictions)*.* The certificate does not replace *Forensic Predictions*. It packages *Forensic Predictions*-style admissibility into a theorem-facing object. This is the point where the operational criterion becomes usable rather than merely definitional.

38 Operational Geometry of $\mathrm{Q}_{\mathrm{op}}(S)$

Once $\mathrm{Q}_{\mathrm{op}}(S)$ is defined as a quotient, one can ask which geometric quantities survive transport and rupture. This matters because the criterion is not only about set membership; it is also about the structural shape of admissible operational differentiation.

Definition 38.1 (Class diameter and separation)**.** For $q \in \mathrm{Q}_{\mathrm{op}}(S)$ define its diameter

$$\mathrm{diam}(q) = \sup_{x,y \in q} \mathrm{dist}\big(h_{S,T}(x; u_\bullet), h_{S,T}(y; u_\bullet)\big)$$

over admissible windows and schedules. For two distinct classes $q, q' \in \mathrm{Q}_{\mathrm{op}}(S)$ define the inter-class separation

$$\mathrm{sep}(q, q') = \inf_{x \in q,\ y \in q'} \mathrm{dist}\big(h_{S,T}(x; u_\bullet), h_{S,T}(y; u_\bullet)\big).$$

Definition 38.2 (Persistence radius)**.** The persistence radius of a class $q \in \mathrm{Q}_{\mathrm{op}}(S)$ is the largest $\rho_q \geq 0$ such that for every admissible trajectory segment starting in q, the decoder-certified class assignment remains inside q on windows of length at least $\tau_{\min}(S)$ under all perturbations of size at most ρ_q that do not target a critical invariant.

Proposition 38.3 (Positive geometry under the criterion)**.** *If* $S \in \mathrm{Consciousness}_{\mathrm{op}}$*, then:*

1. *every class in* $\mathrm{Q}_{\mathrm{op}}(S)$ *has finite diameter on admissible windows;*
2. *there exists at least one pair of distinct classes with positive inter-class separation;*

3. every class has non-zero persistence radius.

Proof. Finite diameter follows from compact admissible support and continuity of admissible histories. Positive separation follows from $I_{\text{diff}}(S) = 1$ and the class-separability clause of $I_{\text{leg}}(S)$. Non-zero persistence radius follows from the persistence-window and noise-robustness clauses in Definition 29.6. □

Proposition 38.4 (Exact equivalence preserves operational geometry)**.** *If $S_1 \simeq_\Gamma S_2$, then class diameter, inter-class separation, and persistence radius are preserved by the induced bijection $\widehat{\chi}$ up to the bi-Lipschitz constants of χ and χ^{-1}.*

Proof. Exact equivalence preserves admissible histories and decoder certification under transport. Bi-Lipschitz bounds imply that history distances are distorted only within the corresponding Lipschitz constants. Therefore diameters, separations, and persistence radii are preserved up to those constants. □

Corollary 38.5 (Rupture contracts geometry)**.** *If a rupture eliminates either I_{diff} or I_{leg}, then the geometric content of $\mathrm{Q}_{\text{op}}(S)$ contracts to triviality or becomes operationally unavailable.*

Proof. If I_{diff} fails, positive separation disappears. If I_{leg} fails, persistence radii and certified class geometry cease to be available under admissible decoding. Either way the geometric structure that underwrites $\mathrm{Q}_{\text{op}}(S)$ is lost. □

39 Intervention Geometry and Class Transition Semantics

Predictions about invariant-targeted interventions should not remain informal. This section encodes them as transition statements on the operational qualia classes themselves.

Definition 39.1 (Class transition operator)**.** Fix an admissible system $S \in \text{Consciousness}_{\text{op}}$, a certified decoder family, and an admissible horizon $T \geq \tau_{\min}(S)$. For an admissible intervention $u \in U$ define the class transition operator

$$\mathsf{T}_u : \mathrm{Q}_{\text{op}}(S) \to \mathrm{Q}_{\text{op}}(S) \cup \{\perp\}$$

by the rule

$$\mathsf{T}_u(q) = q'$$

whenever every admissible representative $x \in q$ is transported by the u-intervened dynamics into histories classified in the same decoder-certified class q'. If no such common class exists, set $\mathsf{T}_u(q) = \perp$.

Definition 39.2 (Critical and sham interventions)**.** An intervention $u \in U$ is *critical* for invariant I_k if it is designed to change the local witness margin $\delta_k(S)$ while leaving the remaining intervention protocol admissible. An intervention is *sham* if it matches the delivery envelope of a critical intervention but is not predicted to alter any critical invariant.

Proposition 39.3 (Well-definedness of class transitions under certification)**.** *Let $S \in \text{Consciousness}_{\text{op}}$. If $u \in U$ is either certified critical or sham and remains within the admissibility envelope of Forensic Predictions, then T_u is well-defined on every class $q \in \mathrm{Q}_{\text{op}}(S)$.*

Proof. Because $S \in \text{Consciousness}_{\text{op}}$, the decoder family is certified and class assignments are stable on admissible windows. If u remains within the admissibility envelope, persistence and continuity keep all intervened histories on admissible support, while legibility prevents class assignment from depending on the choice of representative inside a fixed class. Therefore every class either transports coherently to a unique target class or exits the certified regime, in which case the sink value $\perp$ applies. □

Proposition 39.4 (Targeted invariant loss induces class exit or contraction)**.** *Let* $q \in \mathrm{Q}_{\mathrm{op}}(S)$ *and let* u *be a certified critical intervention for invariant* I_k*. If applying* u *drives the corresponding witness margin to zero, then one of the following occurs:*

1. $\mathsf{T}_u(q) = \perp$*;*
2. $\mathsf{T}_u(q) = q'$ *for some class* q' *whose separation or persistence radius is strictly smaller than that of* q*;*
3. *the entire post-intervention system exits* Consciousness$_{\mathrm{op}}$*.*

Proof. Driving $\delta_k(S)$ to zero triggers the corresponding rupture mode from Section 34. If the rupture destroys admissible support, decoder certification, or non-trivial class structure, then the class leaves the certified domain and $\mathsf{T}_u(q) = \perp$. If a non-trivial class survives, Corollary 38.5 implies contraction of the associated geometry. If the rupture is global, Theorem 33.5 excludes the post-intervention system from Consciousness$_{\mathrm{op}}$. □

Corollary 39.5 (Sham stability)**.** *Let* $u \in U$ *be sham in the sense of Definition 39.2. If* u *stays below all six witness margins, then* $\mathsf{T}_u(q) = q$ *for every* $q \in \mathrm{Q}_{\mathrm{op}}(S)$*.*

Proof. A sham intervention that remains below all witness margins does not cross any rupture threshold. Approximate stability and legibility therefore preserve class membership on every admissible window. □

Remark 39.6 (Why transition semantics matter)*.* This section turns the falsation grammar into a structured intervention calculus. The criterion is therefore not only classificatory; it is also dynamically discriminative in the sense required by *Forensic Predictions*.

40 Baseline Families and Structural Exclusion

A criterion of substrate-invariant operational consciousness matters only if it separates itself from weaker rival families rather than merely renaming them. This section therefore makes the comparison explicit.

Definition 40.1 (Rival baseline families)**.** Fix thresholds $\theta_X, \theta_\Phi, \theta_C > 0$ and an external-behavior tolerance $\eta \geq 0$.

1. The *complexity-only family* $\mathcal{B}_{\mathrm{cx}}(\theta_X)$ contains all admissible systems with complexity surrogate above threshold, for example $\dim(X) \geq \theta_X$.
2. The *computation-only family* $\mathcal{B}_{\mathrm{cmp}}(\theta_\Phi)$ contains all admissible systems whose transition family exceeds a chosen throughput or expressive-capacity threshold.
3. The *connectivity-only family* $\mathcal{B}_{\mathrm{conn}}(\theta_C)$ contains all admissible systems whose effective causal graph exceeds the chosen density or degree threshold.
4. The *weak functional-similarity family* $\mathcal{B}_{\mathrm{fun}}(\eta)$ contains all admissible systems whose external benchmark readouts match within tolerance η, irrespective of internal invariant preservation.
5. The *substrate-privilege family* $\mathcal{B}_{\mathrm{sub}}$ contains all systems declared admissible solely by belonging to a privileged substrate class, typically biological.

Proposition 40.2 (Complexity, computation, and connectivity are not sufficient)**.** *For every choice of thresholds* $\theta_X, \theta_\Phi, \theta_C$*, the families* $\mathcal{B}_{\mathrm{cx}}(\theta_X)$*,* $\mathcal{B}_{\mathrm{cmp}}(\theta_\Phi)$*, and* $\mathcal{B}_{\mathrm{conn}}(\theta_C)$ *are each insufficient for membership in* Consciousness$_{\mathrm{op}}$*.*

Proof. This is a direct sharpening of Proposition 33.6. The definition of $\text{Consciousness}_{\text{op}}$ requires all six invariants. None of the three surrogate families encodes rigid identity, continuity, differentiation, or legibility. Therefore systems may exceed the chosen surrogate threshold while still failing $\text{Consciousness}_{\text{op}}$. □

Proposition 40.3 (Substrate privilege is too strong)**.** *On any admissible universe containing one exact cross-substrate equivalent pair, $\mathcal{B}_{\text{sub}}$ is strictly narrower than* $\text{Consciousness}_{\text{op}}$*.*

Proof. By Corollary 33.3, an exact cross-substrate equivalent pair belongs to the same operational consciousness class whenever the six invariants are preserved. A substrate-privilege family that excludes one member of that pair therefore excludes a valid $\text{Consciousness}_{\text{op}}$ member. □

Proposition 40.4 (Weak functional similarity is too weak)**.** *On any admissible universe containing a same-output rupture pair, $\mathcal{B}_{\text{fun}}(\eta)$ is strictly broader than* $\text{Consciousness}_{\text{op}}$*.*

Proof. A same-output rupture pair is a pair of admissible systems whose external benchmark readouts agree within tolerance η while one member loses at least one critical invariant. Such a pair lies in $\mathcal{B}_{\text{fun}}(\eta)$ by definition, but the ruptured member is excluded from $\text{Consciousness}_{\text{op}}$ by Theorem 33.5. Therefore weak functional similarity is too weak to characterize the present class. □

Theorem 40.5 (The structural criterion is strictly intermediate)**.** *Let $\mathfrak{U}$ be an admissible universe containing:*

1. *at least one exact cross-substrate equivalent pair;*
2. *at least one same-output rupture pair;*
3. *at least one single-rupture witness for each invariant.*

Then $\text{Consciousness}_{\text{op}}$ *is strictly intermediate between substrate-privilege baselines and weak substrate-blind surrogates: it is broader than $\mathcal{B}_{\text{sub}}$, narrower than $\mathcal{B}_{\text{fun}}(\eta)$, and extensionally distinct from the complexity-only, computation-only, and connectivity-only families.*

Proof. Broader-than-substrate-privilege follows from Proposition 40.3. Narrower-than-weak-functional-similarity follows from Proposition 40.4. Distinctness from the surrogate families follows from Proposition 40.2 together with Corollary 35.5. Hence the present criterion is neither a substrate fetish nor a substrate-blind similarity slogan; it is an invariant-preserving structural criterion. □

Remark 40.6 (Scientific value of the comparison)*.* The point of Theorem 40.5 is not rhetorical positioning. It is to show that the criterion makes a genuinely discriminative contribution. A new criterion that cannot separate itself from obvious rivals is not a theorem-level addition to the corpus.

Part VIII
Claim Boundary and Falsation Grammar

41 Forensic Admissibility and Integrity Grammar

Definition 41.1 (Forensic Packet)**.** Each operational run emits a packet

$$K = (\text{tick}, \text{seed}, \text{versionHash}, H_v, \text{metrics}, \text{sensorSummary}, \text{checksum}).$$

Assumption 41.2 (Packet Admissibility)**.** A run is admissible only if checksum, schema, seed consistency, version hash, and H_v consistency all pass.

Definition 41.3 (Severe Violation)**.** A run has a severe violation if any of the following holds: checksum mismatch, schema failure, seed/version inconsistency, or critical-state replay mismatch.

Criterion 41.4 (Integrity Gate)**.** Severe runs are excluded from support counting and marked as hard failures.

Proposition 41.5 (Integrity-First Inference)**.** *If admissibility fails, inferential significance is non-actionable.*

42 Certification, Equivalence, and Rupture

Definition 42.1 (Operational Certification)**.** For an admissible system S, define

$$\mathrm{Cert}(S) = 1 \iff I_{\mathrm{per}}(S), I_{\mathrm{ri}}(S), I_{\mathrm{int}}(S), I_{\mathrm{cont}}(S), I_{\mathrm{diff}}(S), I_{\mathrm{leg}}(S) > 0$$

on a non-empty admissible support under the governance and admissibility constraints inherited from the forensic layer.

Definition 42.2 (Equivalence Discriminator)**.** For two admissible systems S_1, S_2, define

$$\mathrm{Eq}_{\Gamma,\varepsilon}(S_1, S_2) = 1$$

when the relevant signatures match, the admissibility bundle is preserved, and the tolerated invariant deltas remain within the allowed comparison regime.

Definition 42.3 (Rupture Event)**.** For a targeted invariant family J, define

$$\mathrm{Rupt}_J(S) = 1$$

when an admissible intervention or perturbation drives at least one invariant in J below its certification margin so that class membership becomes invalidated, destroyed, or undefined.

43 Prediction and Falsation Grammar

Proposition 43.1 (Complexity-Only Negative Control Prediction)**.** *If a candidate system exhibits high connectivity, high entropy, or rich activity while failing one or more critical invariants, then*

$$\mathrm{Cert}(S) = 0, \qquad S \notin \mathrm{Consciousness}_{\mathrm{op}} \textit{ in certified form}, \qquad \mathrm{Q}_{\mathrm{op}}(S) \textit{ is not certified.}$$

Proposition 43.2 (Ablation Prediction)**.** *If a system is initially certified and a targeted admissible intervention destroys one or more invariants, then the framework predicts loss of certification or transition to an ambiguity-boundary regime.*

Proposition 43.3 (Substrate-Preserved Equivalence Prediction)**.** *If two realizations preserve the relevant signatures, admissibility conditions, and six critical invariants up to the allowed comparison regime, then the framework predicts preserved operational class membership independently of substrate labels.*

Proposition 43.4 (Threshold-Sensitive Boundary Behavior)**.** *The framework predicts stable pass/fail regions together with narrow transition bands near critical margins; it predicts neither global threshold collapse nor perfectly flat insensitivity.*

Proposition 43.5 (Legibility Dependence Prediction)**.** *Operational class membership becomes inadmissible if readouts cease to be separable, noise-robust, intervention-sensitive, or compressible without total collapse.*

Proposition 43.6 (Continuity and Non-Nullity Relevance)**.** *Continuity and non-null differentiation are constitutive. Systems that preserve rich activity while losing continuity or collapsing toward effective null structure fail the operational criterion or leave it undefined.*

Remark 43.7 (What stays outside the core)*.* Cycle-specific status classes, internal support ledgers, and release-facing campaign summaries remain downstream artifacts. The core absorbs only the falsation grammar and claim-boundary semantics, not the mutable empirical status bookkeeping.

44 Claim Boundary Ledger

44.1 What the Core Proves

1. Contractive-projection existence, uniqueness, attractor structure, compactness, and non-collapse under explicit hypotheses.
2. Inverse-limit identity, rigidity, uniqueness, non-duplicability, and computational non-simulability pressure under the absorbed Paper 1 burden.
3. Regime continuity, anti-fragmentation, dichotomy, and null-regime exclusion under the absorbed Papers 2-3 burden.
4. Operational criterion structure, exact/approximate structural equivalence, and rupture logic under the absorbed Paper 5 burden.
5. Admissibility and falsation grammar sufficient to state what would count against the framework without confusing runtime support with theorem closure.

44.2 What the Core Does Not Prove

1. Human phenomenal consciousness.
2. Metaphysical subjectivity.
3. Human-machine equivalence.
4. Biological privilege as a primitive constitutive variable.
5. External validation or public claim closure.
6. Bridge admissibility or any Paper 9 phenomenal-bridge conclusion.

45 Dependency Ledger

1. The foundational Hilbert-space dynamics remain the base layer.
2. Paper 1 content is absorbed as an identity and rigidity extension of that base.
3. Paper 2 content is absorbed as continuity, fragmentation, and regime-constraint structure.
4. Paper 3 content is absorbed as null-regime closure and forced non-nullity.
5. Paper 4 contributes admissibility and integrity grammar.

6. Paper 5 contributes the six-invariant operational criterion and structural-equivalence machinery.
7. Paper 6 contributes the falsation grammar and prediction/failure semantics.
8. Papers 7-9 remain downstream and may depend on this core, but this core does not depend on them.

46 Novelty Ledger

1. The canonical core is now an autonomous modular source tree rather than a monolithic file with external bibliography dependence.
2. Theorems and definitions previously distributed across Papers 1-6 are selectively re-owned inside the core.
3. Claim-boundary, admissibility, and falsation semantics are now explicit inside the source-of-truth tree.
4. Downstream boundaries are frozen in-document rather than left implicit.

47 Redundancy Ledger

1. Identity as inverse limit is owned here as core theorem-level structure rather than split ambiguously across multiple papers.
2. Ontological mass is treated as a canonical invariant with one ownership path.
3. Phenomenological space and compatibility logic are normalized into a single canonical chain for regime results.
4. Cycle-specific empirical status tables remain outside the theorem core and are not duplicated here.

48 Open Assumptions Inventory

1. All formal results remain conditional on their stated hypotheses and admissibility assumptions.
2. Physical realization of CCR-like systems remains open.
3. Empirical instantiation of the operational criterion remains open.
4. External validation remains open.
5. Papers 7-9 introduce downstream burdens not settled by this core.

49 Downstream Boundary Note

Papers 7, 8, and 9 remain downstream of this source tree. Their classificatory, indexed-subjectivity, and bridge-burden layers may cite this core, but they are not part of the autonomous mathematical base rebuilt here.

A Counterexamples for Necessity

A.1 Without H1 (Convexity)

Let $\mathcal{I} = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ (unit circle, not convex).

For $\Psi = (0, 0)$, every point on $\mathcal{I}$ is equidistant. Projection is not unique.

A.2 Without H2 ($\|K\| < 1$)

Let K be a rotation by angle $\theta \neq 0$ with $\|K\| = 1$.

Orbits cycle indefinitely; no fixed point exists.

A.3 Without H3 (Completeness)

Let $\mathcal{H} = C([0, 1], \mathbb{Q})$ (rational-valued continuous functions).

Cauchy sequences may converge to irrational functions outside the space.

A.4 Without H4 (Compactness)

Let $\mathcal{U} = (0, 1)$ (open interval) and $\Gamma(u) = 1/u$.

As $u \to 0$, attractors escape to infinity; $\mathcal{A}^*$ is not compact.

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